

Internet Appendix:

Forward Return Expectations*

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A Proofs

Proof of Proposition 1. Given that $M_{t,t+n}R_{t,t+n} = 1$ by assumption, $\mathcal{C}_t^{(n)} = 0$ in (4). The stated results then follow immediately. $\square$

Proof of Proposition 2. First consider $n = m = 1$, and write

$$\begin{aligned}\varepsilon_{t+1}^{(1)} &= \mu_{t+1}^{(1)} - f_t^{(1)} = \mathcal{L}_{t+1}^{(1)} - \mathcal{L}_t^{(2)} + \mathcal{L}_t^{(1)} + \text{cov}_t(MR_{t,t+2}, r_{t,t+2}) \\ &\quad - \text{cov}_t(MR_{t,t+1}, r_{t,t+1}) - \text{cov}_{t+1}(MR_{t+1,t+2}, r_{t+1,t+2}).\end{aligned}\quad (\text{A1})$$

Consider the first covariance term. Given the joint log-normality of the SDF and returns (and the normality of $r_{t,t+n}$), Stein's lemma gives that

$$\begin{aligned}\text{cov}_t(MR_{t,t+2}, r_{t,t+2}) &= \text{cov}_t(MR_{t,t+1}MR_{t+1,t+2}, r_{t,t+2}) \\ &= \text{cov}_t(mr_{t,t+1} + mr_{t+1,t+2}, r_{t,t+2}) \mathbb{E}_t[MR_{t,t+2}] \\ &= \text{cov}_t(mr_{t,t+1} + mr_{t+1,t+2}, r_{t,t+1} + r_{t+1,t+2}),\end{aligned}$$

where $mr_{t,t+n} = \ln(MR_{t,t+n})$, and where the last line uses that $\mathbb{E}_t[MR_{t,t+2}] = 1$. Having separated the two MR terms, apply Stein's lemma again to obtain

$$\begin{aligned}\text{cov}_t(MR_{t,t+2}, r_{t,t+2}) &= \text{cov}_t(mr_{t,t+2}, r_{t,t+1}) + \text{cov}_t(mr_{t,t+1}, r_{t+1,t+2}) + \text{cov}_t(mr_{t+1,t+2}, r_{t+1,t+2}) \\ &= \text{cov}_t(MR_{t,t+2}, r_{t,t+1}) + \text{cov}_t(MR_{t,t+1}, r_{t+1,t+2}) \\ &\quad + \text{cov}_t(MR_{t+1,t+2}, r_{t+1,t+2}).\end{aligned}\quad (\text{A2})$$

For the first two terms in (A2), by the law of total covariance and using that $\mathbb{E}_{t+1}[MR_{t+1,t+2}] = 1$,

$$\begin{aligned}\text{cov}_t(MR_{t,t+2}, r_{t,t+1}) &= \mathbb{E}_t[MR_{t,t+1}r_{t,t+1} \text{cov}_{t+1}(MR_{t+1,t+2}, 1)] \\ &\quad + \text{cov}_t(MR_{t,t+1} \mathbb{E}_{t+1}[MR_{t+1,t+2}], r_{t,t+1}) \\ &= \text{cov}_t(MR_{t,t+1}, r_{t,t+1}),\end{aligned}\quad (\text{A3})$$

$$\begin{aligned}\text{cov}_t(MR_{t,t+1}, r_{t+1,t+2}) &= \mathbb{E}_t[MR_{t,t+1} \text{cov}_{t+1}(1, r_{t+1,t+2})] + \text{cov}_t(MR_{t,t+1}, \mathbb{E}_{t+1}[r_{t+1,t+2}]) \\ &= \text{cov}_t(MR_{t,t+1}, \mathbb{E}_{t+1}[r_{t+1,t+2}]).\end{aligned}\quad (\text{A4})$$

Turning now to the last term in (A1), the law of total covariance can similarly be applied to obtain that as of time t ,

$$\mathbb{E}_t[\text{cov}_{t+1}(MR_{t+1,t+2}, r_{t+1,t+2})] = \text{cov}_t(MR_{t+1,t+2}, r_{t+1,t+2}).\quad (\text{A5})$$

Taking expectations in (A1), substituting in results (A2)–(A5), and applying the definition of $\widehat{\varepsilon}_{t+1}^{(1)}$, we obtain:

$$\mathbb{E}_t[\varepsilon_{t+1}^{(1)}] = \mathbb{E}_t[\widehat{\varepsilon}_{t+1}^{(1)}] + \text{cov}_t(MR_{t,t+1}, \mathbb{E}_{t+1}[r_{t+1,t+2}]).\quad (\text{A6})$$

Rearranging to solve for $\mathbb{E}_t[\widehat{\varepsilon}_{t+1}^{(1)}]$ yields the stated result for the $n = m = 1$ case. While this case is convenient for straightforward derivations, note that all the above steps apply when using $t + n$ in place of $t + 1$ and using $t + n + m$ in place of $t + 2$, so the stated result holds for general n, m . $\square$

Proof of Proposition 3. Starting again with (A1) and expanding the first covariance term,

$$\text{cov}_t(MR_{t,t+2}, r_{t,t+2}) = \text{cov}_t(MR_{t,t+1}MR_{t+1,t+2}, r_{t,t+1}) + \text{cov}_t(MR_{t,t+1}MR_{t+1,t+2}, r_{t+1,t+2}). \quad (\text{A7})$$

We consider each of the two terms on the right side of (A7) in turn, and in both cases apply the law of total covariance. For the first term, as in (A3),

$$\text{cov}_t(MR_{t,t+1}MR_{t+1,t+2}, r_{t,t+1}) = \text{cov}_t(MR_{t,t+1}, r_{t,t+1}). \quad (\text{A8})$$

For the second term,

$$\begin{aligned} \text{cov}_t(MR_{t,t+1}MR_{t+1,t+2}, r_{t+1,t+2}) &= \mathbb{E}_t[MR_{t,t+1} \text{cov}_{t+1}(MR_{t+1,t+2}, r_{t+1,t+2})] \\ &\quad + \text{cov}_t(MR_{t,t+1}, \mathbb{E}_{t+1}[r_{t+1,t+2}]). \end{aligned} \quad (\text{A9})$$

Using (A8) and (A9) in (A1), applying the definition of $\widehat{\varepsilon}_{t+1}^{(1)}$, and taking expectations,

$$\begin{aligned} \mathbb{E}_t[\varepsilon_{t+1}^{(1)}] &= \mathbb{E}_t[\widehat{\varepsilon}_{t+1}^{(1)}] + \mathbb{E}_t[(MR_{t,t+1} - 1) \text{cov}_{t+1}(MR_{t+1,t+2}, r_{t+1,t+2})] \\ &\quad + \text{cov}_t(MR_{t,t+1}, \mathbb{E}_{t+1}[r_{t+1,t+2}]) \\ &= \mathbb{E}_t[\widehat{\varepsilon}_{t+1}^{(1)}] + \text{cov}_t(MR_{t,t+1}, \mathbb{E}_{t+1}[r_{t+1,t+2}] + \text{cov}_{t+1}(MR_{t+1,t+2}, r_{t+1,t+2})). \end{aligned} \quad (\text{A10})$$

Note from (4) that $\mathcal{L}_{t+1}^{(1)} = \mathbb{E}_{t+1}[r_{t+1,t+2}] + \text{cov}_{t+1}(MR_{t+1,t+2}, r_{t+1,t+2})$. Using this in (A10),

$$\mathbb{E}_t[\widehat{\varepsilon}_{t+1}^{(1)}] = \mathbb{E}_t[\varepsilon_{t+1}^{(1)}] - \text{cov}_t(MR_{t,t+1}, \mathcal{L}_{t+1}^{(1)}).$$

The above steps again apply when using $t+n$ in place of $t+1$ and using $t+n+m$ in place of $t+2$, completing the proof. $\square$

Proof of Lemma A1. To compute the risk-neutral expectation of $g(P_T) = R^\alpha (\ln R)^\beta$, we apply the Carr and Madan (1998) formula. Under standard regularity conditions, we have

$$\begin{aligned} \frac{1}{R_f} \mathbb{E}_t^*[R^\alpha (\ln R)^\beta] &= [g(\bar{P}) - \bar{P}g'(\bar{P})] \frac{1}{R_f} + g'(\bar{P}) P_t \\ &\quad + \int_0^{\bar{P}} g''(K) \text{put}(K) dK + \int_{\bar{P}}^\infty g''(K) \text{call}(K) dK, \end{aligned}$$

where $g'(P) \equiv \frac{\partial g}{\partial P}$ and $g''(P_T) \equiv \frac{\partial^2 g}{\partial P^2}$. The result follows by setting $\bar{P} = F_t^{(n)}$ and simplifying. $\square$

Proof of Proposition A1. (A15) is immediate from Martin (2017) Result 8. (A16) and (A17) follow from Lemma A1 by setting the appropriate α and β and simplifying. $\square$

B Related Literature

This appendix expands upon the literature review in the introduction.

We build on literatures studying the behavior of investor expectations measured both from derivatives prices and in survey data, as discussed briefly in the introduction and in Table 1. For the derivatives-based measures, our paper also builds on Giglio and Kelly (2018), who document excess volatility in long-maturity relative to short-maturity claim prices across a range of term structures.

They focus on the term structure of at-the-money implied variance when considering equity markets. We use a different measure of implied volatility, building on [Gao and Martin \(2021\)](#), and we show how this measure can be connected to the term structure of expected equity returns and forecast errors. Our paper thus connects to other work on the term-structure consistency of implied volatility, including [Stein \(1989\)](#) and [Mixon \(2007\)](#). Our result that average forecast errors are close to zero echoes the finding of [Dew-Becker, Giglio, Le, and Rodriguez \(2017\)](#) that it is costless on average to hedge forward variance news.

As in Table 1, our option-based estimates are also closely related to [Martin \(2017\)](#), who studies option-implied spot expected returns under a covariance condition we build on, and also [Augenblick and Lazarus \(2026\)](#), who derive volatility bounds for fixed-maturity index options. We differ from these papers in our focus on the term structure of expected returns. This brings us slightly closer to [Augenblick et al. \(2025\)](#), who study how updating differs by horizon. For other recent work on measuring expectations from options, see, for example, [Ait-Sahalia, Karaman, and Mancini \(2020\)](#) and [Chabi-Yo and Loudis \(2020\)](#). We also, in contrast with much of this past work, consider option-based measures jointly with survey-based measures of expectations.

[Adam and Nagel \(2023\)](#) provide a review of the literature on survey-based expectations. This literature often finds that investors exhibit less than fully rational expectations in their survey responses, though as above, this behavior varies across settings. In addition to the references in Table 1, [Couts, Gonçalves, and Loudis \(2023\)](#), [Boutros et al. \(2025\)](#), [Gormsen and Huber \(2025\)](#), and [Dahlquist and Ibert \(2026\)](#) provide further recent evidence.¹ We consider how investors expect future expected returns to evolve, whereas past work has largely focused on contemporaneous expected returns. (We note that [De la O and Myers 2021](#) do make use of a forward expectation for CFOs, which they call $\mathbb{E}[r_{t+2,t+10}]$, in their Section 4; we build on this by studying forward expectations for a range of investors, and by studying forecast errors and their implications.) We also provide novel evidence of the consistency of expectations dynamics across settings when considering these forward expectations. Further, to the degree that there remain differences in the behavior of survey expectations across different groups of investors, one advantage of our option-based analysis is that we directly study the expectations expressed in equilibrium behavior.

Finally, in comparing forward and realized spot rates, our analysis appears similar in spirit to tests of the expectations hypothesis (EH) for the fixed-income term structure. Aside from focusing on equity, our methodology is geared towards estimating physical expectations (and particularly forecast errors) for future expected returns; by contrast, the spot and forward rates used in fixed-income EH tests embed risk premia by design, and violations of the EH in that setting are equivalent to bond-return predictability.²

C Measurement Details

C.1 Options Data

U.S. Data. For the 1996 to 2021 period, we obtain end-of-day option prices, index prices, projected dividend yields, and risk-free rates from OptionMetrics. To maximize the sample size, we use options with non-standard expirations (weekly) and settlement (PM). We use the bid/ask midpoint as the

¹Separately, [Schmidt-Engelbertz and Vasudevan \(2025\)](#) study higher-order beliefs (i.e., beliefs about other investors’ beliefs) about returns, finding that these may explain short- and long-horizon expected return variation in a manner consistent with our results.

²Bond-return predictability may, however, arise from the dynamics of physical expectations, as in [Farmer, Nakamura, and Steinsson \(2024\)](#) (see also [d’Arienzo 2020](#)).

option price in the main analysis. We linearly interpolate the risk-free rate curve to match option maturities. If either the dividend yield or risk-free rate is missing, we use the last non-missing observation.

For the 1990 to 1995 period, we obtain intraday option prices from CBOE Market Data Replay, as in [Culp, Nozawa, and Veronesi \(2018\)](#). We obtain end-of-day index prices/returns from CRSP and estimate dividend yields from lagged one-year cum/ex-dividend index returns. We obtain Treasury bill rates and constant maturity Treasury yields from FRED to construct risk-free rates, again following Culp, Nozawa, and Veronesi.³

Unlike OptionMetrics, CBOE provides intraday quotes. To construct end-of-day prices, we first apply filters to the intraday data and then use the last available quote. We drop quotes with the missing codes of 998 or 999. We drop quotes with negative bid-ask spreads. We correct erroneously recorded quotes — quotes with strike price less than 100 — by multiplying the strike/option price by 10. We drop end-of-day quotes that increase and then decrease fourfold (or vice versa), following similar filters in [Andersen, Bondarenko, and Gonzalez-Perez \(2015\)](#) and [Duarte, Jones, and Wang \(2024\)](#). We interpret these large reversals as probable data errors. To validate these filters, we compare data from CBOE and OptionMetrics in 1996. We match approximately 99.3% of option prices in OptionMetrics, suggesting these filters are not unreasonable.

We apply standard filters to the end-of-day data, as in [Constantinides, Jackwerth, and Savov \(2013\)](#), among others. (1) We drop options with special settlement. (2) To eliminate duplicate quotes, we select the quote with highest open interest. (3) We drop options with fewer than seven days-to-maturity. (4) We drop options with price less than the minimum tick size.⁴ (5) We drop options with zero bid prices or negative bid-ask spreads. (6) We drop options that violate static no-arbitrage bounds:

$$\text{put}(K) \leq Ke^{-r\tau} \quad \text{call}(K) \leq P_t.$$

(7) We drop options for which the [Black-Scholes](#) implied volatility computation does not converge and options with implied volatility less than 5% or greater than 100%.

Global Data. We again obtain end-of-day option prices, index prices, projected dividend yields, and risk-free rates from OptionMetrics. Unlike the U.S. data, most option prices are either end-of-day settlement prices or last traded prices. Only a small fraction are from either bid/ask prices. The index price is time synchronized to the option price. If the index price is missing, we obtain the end-of-day price from Compustat Global. Risk-free rates are from currency-matched LIBOR curves. Dividend yields are from put-call parity and so are maturity-specific. As before with risk-free rates, we linearly interpolate the dividend yield curve to match option maturities. We apply the same filters to the end-of-day data as with the U.S. data, except for filters that require bid/ask prices.

Table [A1](#) describes the global sample. The Europe and pan-Europe sample begins in January 2002 and ends in September 2021. The Asia-Pacific sample begins in January 2004 and ends in April 2021. Our global sample closely follows [Kelly, Pástor, and Veronesi \(2016\)](#) and [Dew-Becker and Giglio \(2023\)](#), but we also use pan-European Stoxx indexes. These indexes represent a substantive addition to the sample. At long maturities, the Euro Stoxx 50 is arguably the most liquid options market in the world, as is the case with the dividend futures market.⁵

Main Sample. As the global data is not equally robust across exchanges, we select the most reliable exchanges for the main analysis. We select the main sample by elimination. First, we drop Belgium,

³The CBOE website is <https://datashop.cboe.com/mdr-quotes-trades-data>.

⁴For the tick sizes of major index option markets, see [Andersen, Fusari, and Todorov \(2020\)](#).

⁵For more details about the dividend futures market, see [Binsbergen and Koijen \(2017\)](#).

Finland, Netherlands, the Stoxx Europe 50/600, Korea, and Taiwan because they do not consistently have dense option surfaces, as seen in Panel A of Figure A15. Second, we drop Belgium, Korea, and Taiwan because they do not consistently have long-maturity options, as seen in Figure A16. Finally, we drop Finland, the Stoxx Europe 50/600, Sweden, China, and Japan because they do not consistently have deep out-of-the-money options, as seen in Figure A17. After dropping these exchanges, several for multiple reasons, we have 10 exchanges for the main sample, as seen in the last column of Table A1. As a robustness check, we examine the full sample of 20 exchanges in Table A10.

C.2 Option-Based Implementation

Methodology. On each date and separately for puts/calls,

1. We convert option prices to implied volatilities via [Black-Scholes](#). We follow an extensive literature on option-implied risk-neutral densities that finds interpolation/extrapolation more conducive in the space of implied volatilities, not option prices.⁶
2. We fit a Delaunay triangulation to implied volatilities. The grid consists of strike prices between $\underline{K} = 0.10 \times P_t$ and $\bar{K} = 2.00 \times P_t$ with $\Delta K = 0.001 \times P_t$ and maturities $\tau = 30, 60, 91, 122, 152, 182, 273, 365$ days. The triangulation extrapolates as necessary with the nearest implied volatility in simple moneyness and time-to-maturity space.⁷
3. We convert the triangulation of implied volatilities back to option prices via Black-Scholes. We then use the implied triangulation of option prices to evaluate the LVIX integral in (7) via Gaussian quadrature.
4. With the LVIX in hand, we can compute spot rates, forward rates, and forecast errors via Proposition 1, as visualized in Figure A13.
5. We occasionally find negative forward rates. [Gao and Martin \(2021\)](#) argue that negative forward rates are unlikely theoretically and likely represent data errors. We follow Gao and Martin and drop such observations, but our results are not quantitatively sensitive to this choice.

Discussion. Our methodology addresses the main empirical challenges in moment computation: discretization, truncation, and interpolation bias. Discretization and truncation bias relate to the strike dimension of option prices, interpolation bias to the maturity dimension. Our contribution is not to identify these biases — our discussion below closely follows that in [Jiang and Tian \(2007\)](#), among others — but rather to examine the implications for the LVIX.

First, discretization bias arises because (7) requires numerical integration. To address this bias, we integrate on a fine grid of interpolated option prices in step 3. Second, truncation bias arises because (7) requires integration over an infinite range of strikes in theory. In practice, we truncate the integral. To address this bias, we extrapolate and integrate over strikes well beyond the range of observable options in step 2. Finally, interpolation bias arises because (7) typically requires options with unavailable maturities. For example, we do not typically observe options with exactly 182 or 365 days-to-maturity. To address this bias, we interpolate the volatility surface at target maturities in step 2.

⁶For reviews of the literature on risk-neutral densities, see [Figlewski \(2010\)](#) and [Malz \(2014\)](#).

⁷We implement the Delaunay triangulation in Matlab via the `scatteredInterpolant` function.

Our main concern is that the bias varies systematically. One possibility is cross-sectional variation: the bias is larger at longer maturities. If we overestimate long-maturity forward rates, then we would find more negative forecast errors. Another possibility is business-cycle variation: the bias is larger in bad times. If we overestimate forward rates in bad times, then we would find more predictable forecast errors. In either case, measurement error works against the rational expectations null.

We examine the role of measurement error in Appendix D below.

C.3 CFO Survey Data Break

As discussed in Section 3.2, the CFO survey underwent a change in administration in 2020. Prior to 2020, the survey was run by Duke University’s Fuqua School of Business (overseen by John Graham and Campbell Harvey). In spring 2020, it was announced that the survey would be run as a collaborative partnership between Duke, the Federal Reserve Bank of Richmond, and the Federal Reserve Bank of Atlanta; see [Graham et al. \(2020\)](#) for details. The redesign affected the sample composition, the survey design, and the way return expectations are elicited.⁸

These changes induce a discrete level shift in forward return expectations, as shown in Figure A9. The top panel of the figure plots the raw (unadjusted) 1-year, 9-year CFO forward risk premium series. The upward shift in 2020 is clearly evident: the post-2020 CFO sample predicts substantially higher returns unconditionally than the previous sample did, and the pre-to-post-2020 shift swamps the cyclical variation in the entire pre-2020 sample. If left uncorrected, this break would affect estimates of CFO forward-rate cyclicalities (which estimate covariances between the forward rate and cyclical proxies) and any regressions on CFO forward rates, confounding true cyclical movements with the survey redesign itself.

To correct for this structural break, all of our CFO-based regressions and analyses include an indicator variable that equals one from 2020 onward. The bottom panel of Figure A9 plots the CFO forward risk premia after removing the fitted effect of this post-2020 indicator. Once this level shift is taken out, the series appears much more stable and comparable across the full 2001–2025 sample, and the cyclical patterns we document are not driven mechanically by the survey redesign. All regressions with CFO data also drop two observations in 2019 that mix forward rates from before the survey redesign (when the survey was run by Duke Fuqua) with realized spot rates from after the survey redesign (when the survey was run by a collaborative partnership).

C.4 State-Variable Proxies

As discussed in Section 4.3, estimating the cyclicalities of forward rates and forecast errors requires we take a stand on the underlying state variable. Since there is no single empirical measure that captures the state of the economy, we consider a number of state-variable proxies from the literature, which we discuss here.

The baseline set of variables is largely from [Dahlquist and Ibert \(2026\)](#). The first proxy is the earnings yield based on the cyclically-adjusted price-to-earnings ratio or CAPE: the excess CAPE yield for risk premia and CAPE yield for expected returns.⁹ The second proxy is the SVIX index

⁸See the announcement ([Graham et al. 2020](#)) for details on the sample composition. For return expectations, the pre-2020 surveys effectively cued respondents to think in terms of excess returns by providing the current Treasury yield on the same screen as the question about S&P return expectations (e.g., see <https://cfosurvey.fuqua.duke.edu/wp-content/uploads/2019/12/2019-Q4-US-Toplines-1.pdf>); the current surveys no longer provide that information (see https://frbaressurvey.co1.qualtrics.com/jfe/form/SV_23m5JZjAa82BE3j).

⁹For risk premia, we measure the earnings yield here as the excess CAPE yield, defined as the inverse

from [Martin \(2017\)](#): the one-year option-implied risk premium or expected return. This variable can be thought of as the physical expectation of the gross market return from the standpoint of an unconstrained log utility investor fully invested in the market. It is closely related to the LVIX index from [Gao and Martin \(2021\)](#) and the VIX index from the CBOE (see footnote 24 below). The last proxy is an NBER recession indicator. This may be the most intuitive measure of bad times, but its drawback is that it is not available in real time. The NBER Business Cycle Dating Committee typically only identifies recessions after the fact and so this measure suffers from look-ahead bias.

The next set of variables are business-cycle indicators from [Nagel and Xu \(2023\)](#). These include five proxies: industrial production growth; the slope of the Treasury yield curve or the term spread; the difference between Baa and Aaa corporate bond yields or the default spread; the real activity factor from [Ludvigson and Ng \(2009\)](#); and the 30-day CBOE Volatility Index VIX². For more details about measurement, we refer the reader to [Nagel and Xu \(2023\)](#).

The final set of variables are other indicators of financial market conditions, most of which are frequently used in the literature. These include eight proxies: asset manager return expectations from [Dahlquist and Ibert \(2024\)](#); the P/E ratio from [Welch and Goyal \(2008\)](#); the risk aversion index from [Bekaert, Engstrom, and Xu \(2022\)](#); the Chicago Fed National Activity Index (CFNAI); the credit spread index from [Gilchrist and Zakrajšek \(2012\)](#), updated by [Favara et al. \(2016\)](#); the Chicago Fed National Financial Conditions Index (NFCI); the St. Louis Fed Financial Stress Index (STLFSI); and as a summary, the first principal component of the NFCI, the Chicago Fed Adjusted National Financial Conditions Index (ANFCI), and the Kansas City Fed Financial Stress Index (KCFSI).

As stated above, we obtain asset manager return expectations from [Dahlquist and Ibert \(2024\)](#). For risk premia, we (loosely speaking) use the mean expectation from their Table 2, Column (1); for expected returns, we use the mean expectation from Column (3).¹⁰ As their sample of asset managers is highly unbalanced, especially in the beginning of their data, we linearly interpolate as necessary between adjacent observations to generate a continuous time-series. The sample runs from November 1997 to April 2021.¹¹

D Measurement Error in Option-Based Statistics

As introduced in Appendix C.2, our option-based estimation faces the possibility of measurement error. We consider the effect of such measurement error here.

of the CAPE (price to 10-year earnings ratio) less the 10-year real yield. This follows [Shiller, Black, and Jivraj \(2020\)](#), who advocate the use of the excess CAPE yield when considering the equity premium; it also appears more stationary than the CAPE yield. We obtain data on the CAPE yield and excess CAPE yield from Robert Shiller’s [website](#).

¹⁰More specifically, we extract the time fixed effects from a panel regression of risk premia on manager-times-horizon fixed effects and time fixed effects and likewise from a panel regression with expected returns. Unlike their Table 2, neither specification includes the CAPE.

¹¹Most state-variable proxies are available for the entire sample from January 1990 to December 2024. The exceptions are as follows: given our sample of option prices, option-implied return expectations end in December 2021; as discussed in the text above, asset manager return expectations begin in November 1997 and end in April 2021; the STLFSI begins only in January 1994; the KCFSI and thus the FCI/FSI principal component begins only in February 1990.

D.1 Alternative Integration Bounds

Theory. We first analyze truncation bias in theory. To motivate the analysis, we note that the integrand in (7) — the option price divided by the strike price — approaches zero in both integration limits:

$$\lim_{K \rightarrow 0} \frac{\text{put}(K)}{K} = 0 \quad \lim_{K \rightarrow \infty} \frac{\text{call}(K)}{K} = 0.$$

This implies that *all* of the integral’s mass corresponds to observable, near-the-money strikes and *none* corresponds to unobservable, deep out-of-the-money strikes. Figure A14 visualizes this point in the [Black-Scholes](#) model.

Our point above is qualitative: truncation bias might not be large and might even be small. To understand the quantitative implications, we turn to simulations of parametric option pricing models. With knowledge of the true data generating process, we can quantify how methodological choices — truncation of the integral and extrapolation of the implied volatility surface — affect truncation bias. We focus on how the bias varies between short and long maturities as well as good and bad times. Our thought experiment closely follows a similar exercise for the VIX in [Jiang and Tian \(2007\)](#).

Figure A6 reports the results. Panel A truncates the volatility surface. As an example, the 20% truncation bound evaluates the integral from strike $\underline{K} = 0.80 \times P$ to $\overline{K} = 1.20 \times P$. On the left, we consider a [Black-Scholes](#) model. We find a small truncation bias in good times but a large truncation bias in bad times. In good times, volatility is low, deep out-of-the-money options are cheap, and so the bias is small. In contrast, in bad times, volatility is high, deep out-of-the-money options are expensive, and so the bias is large. The bias in forward rates is especially large because longer-maturity option prices have more time value. On the right, we consider a stochastic volatility model with jumps ([SVJ](#)). We again find an uncomfortably large truncation bias. Relative to Black-Scholes, the bias is larger when volatility is low, but smaller when volatility is high because volatility mean reverts in SVJ.¹²

Panel B extrapolates the volatility surface. As an example, the 20% extrapolation bound first truncates the volatility surface at the 20% out-of-the-money strike, then extrapolates the surface with the corresponding implied volatility, and finally evaluates the integral from strike $\underline{K} = 0.10 \times P$ to $\overline{K} = 2.00 \times P$. We continue with a SVJ model. Relative to Panel A, we find that extrapolation reduces truncation bias across the board. More importantly, truncation bias is small in both good and bad times. Our extrapolation scheme reduces, if not eliminates, the cyclical component in truncation bias.

To build intuition as to why extrapolation works, we formally describe the experiment. Let $\text{put}(K, \sigma)$ be the Black-Scholes price of a put with strike K and implied volatility σ . Let $IV_p(K)$ be the true implied volatility of a put with strike K . In the left tail, the experiment replaces true put prices with the approximation:

$$\text{put}(K, IV_p(K)) \approx \text{put}(K, IV_p(0.80 \times P)) \text{ for } K \leq 0.80 \times P.$$

The notation and approximation are analogous for calls. In the right tail, the experiment replaces true call prices with the approximation:

$$\text{call}(K, IV_c(K)) \approx \text{call}(K, IV_c(1.20 \times P)) \text{ for } K \geq 1.20 \times P.$$

The success of this approximation depends on two factors. The first is the wedge between near-the-money and deep out-of-the-money volatility. The second is the sensitivity of Black-Scholes prices to

¹²We compute SVJ prices in Matlab via the `optByBatesNI` function.

volatility. This sensitivity approaches zero in both integration limits:

$$\lim_{K \rightarrow 0} \frac{\partial \text{put}(K, \sigma)}{\partial \sigma} = 0 \quad \lim_{K \rightarrow \infty} \frac{\partial \text{call}(K, \sigma)}{\partial \sigma} = 0.$$

This implies that what matters is not *how* we extrapolate but rather *whether* we extrapolate; that is, the wedge is second-order.

The analysis above suggests that truncation bias leads to underestimates of forward rates, especially in bad times, but that extrapolation can reduce this bias, especially its cyclical component. The analysis is somewhat silent on forecast errors because we did not take a strong stand on the dynamics of the underlying state variable. To make progress on this front, we must turn to the data, which we do so below.

Data. We next analyze truncation bias in the data. To motivate the analysis, we note that there is business-cycle variation in the integral’s support. In bad times, stock prices decrease, and this shifts the distribution of strikes upward. Figure A7 visualizes this point in the U.S. sample. This suggests that truncation bias might be able to generate systematic measurement error, which is required for measurement error to explain forecast-error predictability.¹³

To better understand the relevance of this threat, Table A5 examines alternative integration bounds. To build intuition, the top subpanel artificially truncates the volatility surface. The first row truncates the volatility surface at the 25% out-of-the-money strike. Let K_{min} be the minimum put strike in the data. The lower integration bound is

$$\underline{K} = \max \{K_{min}, 0.75 \times P_t\}.$$

The notation and bound are analogous for calls. Let K_{max} be the maximum call strike in the data. The upper integration bound is

$$\overline{K} = \min \{K_{max}, 1.25 \times P_t\}.$$

In other words, at best, the first row evaluates the integral in (7) from $\underline{K} = 0.75 \times P_t$ to $\overline{K} = 1.25 \times P_t$, but the bounds might be even shallower, depending on the availability of option prices. The next four rows consider deeper but otherwise analogous bounds. The sixth row evaluates the integral over all observable strikes. Each of these bounds naturally varies both by time and maturity with the availability of option prices.

When considering very narrow truncation limits, our results for risk premia are statistically less strong, which is somewhat expected given that this induces severe truncation bias in the approximated integral. More generally, with shallow bounds, forecast errors are relatively large on average but less predictable. With deep bounds, forecast errors are relatively small on average but more predictable. The fact that our results become weaker following the artificial truncation suggests that the limited availability of strikes does not mechanically produce our results — in fact, such limitations in the data would weaken rather than strengthen our results.

As mentioned above, for measurement error (which includes, but is not limited to, truncation bias) to explain the results, we would have to *overestimate* forward rates in bad times. This upward bias would mechanically generate predictably negative forecast errors. But the experiment above suggests that in the presence of measurement error driven by truncation bias, we would likely *underestimate* forward rates in bad times. Figure A8 visualizes this point by showing how the

¹³In contrast, neither discretization nor interpolation bias varies with the business cycle. The spacing of option strikes is typically time-invariant. The availability of option maturities typically varies deterministically.

estimated forward and spot risk premia change under the different integration bounds. Measurement error therefore does not seem to help explain the results, either quantitatively as in the table or even qualitatively, as it works in the wrong direction.

The bottom subpanel extrapolates the volatility surface. The first row considers static integration bounds, following a similar robustness check in [Gormsen and Jensen \(2025\)](#):

$$\left[\underline{K}^{(n)}, \overline{K}^{(n)} \right] = \begin{cases} [0.75, 1.25] \times P_t & n \in \{1, 2\} \\ [0.55, 1.45] \times P_t & n \in \{3, 4, 5\} \\ [0.35, 1.65] \times P_t & n \in \{6, 9\} \\ [0.20, 1.80] \times P_t & n \in \{12\}. \end{cases}$$

These bounds vary by maturity. This variation is logical because deep out-of-the-money options have more time value at longer maturities. The second row considers dynamic integration bounds, again following a similar robustness check in Gormsen and Jensen:

$$\underline{K}^{(n)} = \max \left\{ 0.10, 1.00 - 5\sigma_t^{(n)}\sqrt{\tau} \right\} \times P_t \quad \overline{K}^{(n)} = \min \left\{ 2.00, 1.00 + 5\sigma_t^{(n)}\sqrt{\tau} \right\} \times P_t,$$

where $\sigma_t^{(n)}$, the price of the volatility contract in [Bakshi, Kapadia, and Madan \(2003\)](#), proxies for market volatility:

$$\begin{aligned} \left(\sigma_t^{(n)} \sqrt{\tau} \right)^2 &= \frac{1}{R_f} \mathbb{E}_t^* \left[(\ln R)^2 \right] \\ &= \int_0^{P_t} \frac{2 \left(1 + \ln \left[\frac{P_t}{K} \right] \right)}{K^2} \text{put}(K) dK + \int_{P_t}^{\infty} \frac{2 \left(1 + \ln \left[\frac{P_t}{K} \right] \right)}{K^2} \text{call}(K) dK. \end{aligned} \quad (\text{A11})$$

These bounds vary by time with volatility. This variation is logical because deep out-of-the-money options are more expensive in bad times. The third row considers the baseline integration bounds, as discussed in Appendix C.2. Regression slopes and average errors are quantitatively consistent across all extrapolation schemes, consistent with the idea above that what matters is not how we extrapolate but rather whether we extrapolate. This further helps alleviate concerns about the specific integration bounds used in our baseline: observations very deep in the tails contribute little mass to the LVIX integral,¹⁴ consistent with the theory discussed just above.

D.2 Alternative Measures

The appendix considers robustness checks where we use alternative choices to measure spot rates, forward rates, and forecast errors. Some of these choices are motivated by liquidity. As an example, if liquidity were an issue, then we would expect use of bid/ask prices to significantly alter the results (this is not the case). Other choices are motivated as checks on how exactly we fit a volatility surface to the data. As an example, the SVI surface below explicitly imposes no-arbitrage on the surface (which does not seem to matter either).

Table A6 presents the results. Regression slopes and average errors are quantitatively robust to these choices in all cases, while maintaining their statistical significance in most cases.

¹⁴To see this directly (without the need to extrapolate), one can consider the Euro Stoxx 50, which has a relatively dense grid of deep out-of-the-money options. At the 6-month horizon, the region $1.65 \leq K/P_t \leq 2.00$ contributes at most five basis points to the integral (with an average contribution near zero).

Liquidity Filters. The top subpanel repeats the analysis with additional liquidity filters. The second row considers an outlier filter, following similar filters in [Constantinides, Jackwerth, and Savov \(2013\)](#) and [Beason and Schreindorfer \(2022\)](#). On each date and separately for puts/calls, we first fit a quadratic function to implied volatilities in terms of moneyness K/P and time-to-maturity. To minimize the effect of deep out-of-the-money, short/long-maturity options, we only use options with moneyness $0.65 \leq K/P \leq 1.35$ and maturity $14 \leq \tau \leq 365$ days. We then drop influential observations via Cook’s Distance. The third row considers an open interest filter. We drop options with zero open interest, as seen in Panel B of Figure [A15](#). We do not have open interest data before 1996. The fourth row combines the outlier and open interest filters.

Bid/Ask Prices. The bottom subpanel repeats the analysis with bid/ask prices, following similar robustness checks in [Martin \(2017\)](#) and [Gao and Martin \(2021\)](#). We only have bid/ask prices in the U.S. sample. The first row reports the baseline results with the bid-ask midpoint. The second row repeats the analysis with bid prices, the third ask prices.

Volatility Surface. The second-to-last row in the top subpanel repeats the analysis with the interpolated volatility surface from OptionMetrics. OptionMetrics provides interpolated [Black-Scholes](#) implied volatilities on a constant moneyness/maturity grid.¹⁵

SVI Surface. The last row in the top subpanel repeats the analysis with the stochastic volatility inspired (SVI) surface from Jim Gatheral at Merrill Lynch.¹⁶ Our implementation of the SVI surface closely follows [Berger, Dew-Becker, and Giglio \(2020\)](#) and [Beason and Schreindorfer \(2022\)](#). We parameterize squared [Black-Scholes](#) implied volatilities with the function

$$\sigma_{BS}^2(t, \kappa, \tau) = a + b \left(\rho(\kappa - m) + \sqrt{(\kappa - m)^2 + \sigma^2} \right), \quad (\text{A12})$$

where κ is standardized forward moneyness

$$\kappa = \frac{\ln K - \ln F_t^{(n)}}{\sigma_t^{(1)} \sqrt{\tau}},$$

$\sigma_t^{(1)}$ proxies for the risk-neutral volatility of the market return as in [\(A11\)](#), and each parameter is a linear function of time-to-maturity (e.g., $a = a_0 + a_1\tau$).

On each date, we estimate parameters $\theta = (a_0, a_1, b_0, b_1, \rho_0, \rho_1, m_0, m_1, \sigma_0, \sigma_1)$ that minimize the implied volatility RMSE between the surface [\(A12\)](#) and the data, subject to standard no-arbitrage constraints: option prices are nonnegative and monotonic/convex in K . We check these constraints on a grid with moneyness between $-20 \leq \kappa \leq 0.50$ for puts, between $-0.50 \leq \kappa \leq 10$ for calls, and maturities $\tau = 30, 60, 91, 122, 152, 182, 273, 365$ days. We estimate the surface with outlier-filtered, as discussed above, out-of-the-money puts/calls: puts with $\kappa \leq 0$ and calls with $\kappa \geq 0$. We estimate the surface separately for puts/calls and separately for short/long-maturity options: $14 \leq \tau \leq 122$ days and $122 < \tau \leq 365$ days, respectively.¹⁷

¹⁵The literature often uses this surface for American-exercise options because OptionMetrics reports an equivalent, European-exercise, implied volatility (see, for example, [Kelly, Lustig, and Van Nieuwerburgh 2016](#) and [Martin and Wagner 2019](#)).

¹⁶For more details, see [Gatheral and Jacquier \(2011, 2014\)](#). For a textbook treatment, see [Gatheral \(2011\)](#).

¹⁷For more details about no-arbitrage violations, see [Ait-Sahalia and Duarte \(2003\)](#).

D.3 Additional Robustness Checks

This appendix examines additional empirical exercises related to measurement error. We exploit cross-sectional variation in liquidity across countries and across horizons as well as time-series variation in liquidity in the U.S. sample. We discuss each in turn.

This empirical evidence shows that liquidity-driven measurement error, in fact, weakens our results. This is because illiquidity biases our estimates of forward rates downwards, not upwards as required for measurement error to generate spurious results. This evidence makes us confident that our results on forward rates and forecast errors are driven by the true underlying process of these variables and are very unlikely to be an artifact of measurement error.

Split-Sample Regressions. The first piece of evidence comes from using time variation in the liquidity of option markets in the U.S. sample. Option markets have generally become more liquid over time — on any given trading day, there are literally thousands more options in 2021 than in 1990. More concretely, the surface is more dense, as seen in Figure A18, and the surface extends further into the tails, as seen in Figure A7.

Table A7 reports the results. In the first half, when measurement error is more severe, forecast errors are less predictable. In the second half, when measurement error is less severe, forecast errors are more predictable, even without NBER recessions. That is, our results are stronger in the latter part of our sample. While things other than liquidity have also changed over time, this result similarly suggests that measurement error biases us against our result, and that observing a complete and more liquid set of option prices would strengthen our results.

Alternative Horizons. The next piece of evidence comes from using cross-sectional variation in liquidity across horizons. Short-maturity options (less than 3 months) are substantially more liquid than long-maturity options (more than 3 months), not only in the U.S. sample but also worldwide.

Table A8 considers short horizons: $2 \leq n + m \leq 3$. At these horizons, measurement error is less of a concern. The surface is more dense and extends further into the tails, as seen in Figures A18 and A7. Moreover, the assumptions that the expectations hypothesis holds (see Section 2.2 for more details) and that dividends are known ex ante (see Section 3.1 for more details) are more reasonable at short horizons. Our results continue to hold at shorter horizons. This finding reinforces the idea that measurement error can only weaken our results.

Table A9 considers longer horizons: $4 \leq n + m \leq 12$ (the baseline horizon is $n = m = 6$ in the bottom subpanel). At these horizons, measurement error remains a concern, but the results continue to hold. Holding $n + m$ fixed, forecast errors are relatively small on average and less predictable with small n , relatively large on average and more predictable with large n .

Alternative Samples. The last piece of evidence arises from leveraging cross-country differences in liquidity in the options market. Some countries in our sample have relatively illiquid options markets (like Australia or Spain), while others have very liquid markets (like Germany, the Euro Stoxx 50, and of course, the U.S.). This is especially true when considering the main sample of 10 exchanges versus the full sample of 20 exchanges, as shown in Table A1, which were explicitly categorized on the basis of liquidity. We show that our results are stronger in the most liquid option markets. This finding suggests that if measurement error influences our results, it does so by *weakening* our results and biases us towards finding no effects.

As motivation, Figure A19 visualizes the relationship between spot and forward rates in the global sample, analogous to Figure 2 in the U.S. sample. Table A10 reports the corresponding results. Given liquidity concerns, we make two important changes here relative to the baseline analysis.

First, we focus on horizons substantially shorter than the baseline analysis because, as discussed above, there is even less liquidity at long horizons in the non-U.S. sample and especially in the less-developed option markets considered in this table. So, while liquidity remains a concern in the less-developed markets, it is probably slightly less severe at shorter horizons. Second, we integrate over only observable option prices (that is, we do not extrapolate the surface into the tails as in the baseline analysis). We did not want to spuriously extrapolate the surface and the analysis above in which we consider alternative integration bounds suggests that this is likely to be a conservative approach (that is, bias the results towards the rational expectations null). The table reports results that are quantitatively similar to, but at times somewhat weaker than, those in the main sample. That said, the fact that the results are not stronger in the full sample pushes against the notion that illiquidity in itself can produce rejections of rationality.

As an additional test, we also find that the slopes are substantially weaker — that is, closer to the rational expectations null — in exchanges with more illiquid markets, further suggesting that liquidity-driven measurement error cannot spuriously produce significant results. Unlike the full sample analysis above, we conduct this analysis within the main sample and for the baseline horizon, so this likely speaks better to results as a whole. We briefly outline this additional but unreported robustness check. Our proxy for option liquidity is the share of long-maturity options with positive open interest. The cross-exchange ordering of liquidity is reasonable. The usual suspects, like the SPX, SX5E, and DAX, are the most liquid. With this proxy, we examine the cross-exchange relationship between liquidity and the Mincer-Zarnowitz and error-predictability regression slopes. For both slopes, we find a strong negative correlation. Our takeaway is again that illiquidity works against the results: exchanges with less liquidity have slopes closer to the rational expectations null.

E Risk-Based Explanations for Option-Based Results

This appendix asks what would be required in order for option-based forecast errors to be rationalized by the behavior of the stochastic discount factor alone. The main conclusion is that the required behavior of the SDF is difficult to reconcile with standard models.¹⁸

E.1 Overview

Outside log utility, estimated spot and forward rates partly reflect risk premium adjustments. While these largely cancel out in forecast errors, a risk premium term remains: expected forecast errors are

$$\mathbb{E}_t[\hat{\varepsilon}_{t+n}^{(m)}] = \mathbb{E}_t[\varepsilon_{t+n}^{(m)}] - \varsigma_t, \quad \text{where } \varsigma_t = \begin{cases} \text{cov}_t(MR_{t,t+n}, \mathbb{E}_{t+n} r_{t+n+m}) & (\text{log-normal case}), \\ \text{cov}_t(MR_{t,t+n}, \mathcal{L}_{t+n}^{(m)}) & (\text{general case}), \end{cases}$$

from Propositions 2–3. To simplify analysis, we maintain focus on the $n = m = 1$ case (where one period is 6 months) and suppress the dependence of ς_t on n and m .

As a starting point for the potential role of risk premia, Figure A10 plots the predicted values of the risk-premium forecast errors from the regressions in Table 4. These are empirical estimates of $\mathbb{E}_t[\hat{\varepsilon}_{t+1}^{(1)}]$, and $-\varsigma_t$ must take on these values in order to rationalize our results through risk premia alone (so that true forecast errors satisfy $\mathbb{E}_t[\varepsilon_{t+1}^{(1)}] = 0$). The main challenge is that the covariance term $-\varsigma_t$ must flip sign over time: the covariance term must be significantly negative in bad times,

¹⁸We note that in this and remaining appendix sections, we often streamline formal exposition and notation by using μ , f , and ε to represent objects corresponding to either expected returns or risk premia.

which means it must be positive in good times to remain close to zero on average. We show below that this requires the price of discount rate risk to be highly volatile — close to zero during good times and high during bad times.¹⁹

E.2 Examining the Risk Premium Term

To better understand the properties required of ς_t in order to match Figure A10, it is useful to formalize its relation to a discount-rate risk premium. As in Proposition 2, assume for now that the SDF and returns are jointly log-normal. Then by Stein’s lemma and the fact that $\mathbb{E}_t[MR_{t+1}] = 1$,

$$\text{cov}_t(MR_{t,t+1}, \mathbb{E}_{t+1} r_{t+2}) = \text{cov}_t(m_{t+1} + r_{t+1}, \mathbb{E}_{t+1} r_{t+2}). \quad (\text{A13})$$

For ς_t to flip sign, the covariance in (A13) must thus flip sign. To see how this covariance is tied to discount-rate risk, define $P_{\mathbb{E},t}$ as the price of the claim that pays $-\mathbb{E}_{t+1} r_{t+2}$ next period, and define its ex ante Sharpe ratio (in log-return terms) as $SR_{\mathbb{E},t}$.²⁰ This Sharpe ratio captures the price of risk for exposure to increases in expected returns (or discount rates).

Rewriting (A13), the sign of ς_t is given by

$$\text{Sign}(\varsigma_t) = \text{Sign}\left(SR_{\mathbb{E},t} + \rho_t(r, \mathbb{E}_{t+1} r)\sigma_t(r_{t+1})\right),$$

where $\rho_t(r, \mathbb{E}_{t+1} r) = \text{Corr}(r_{t+1}, \mathbb{E}_{t+1} r_{t+2})$. This correlation is likely negative, as low realized returns are associated with higher expected returns going forward; for simplicity, start by assuming that $\rho_t(r, \mathbb{E}_{t+1} r) = -1 \forall t$.²¹ In this case, the above expression shows that $SR_{\mathbb{E},t}$ must be higher than $\sigma_t(r)$ for ς_t to be positive, and lower than $\sigma_t(r)$ for ς_t to be negative. Since ς_t must be positive in bad times and negative in good times, $SR_{\mathbb{E},t}$ must vary more than $\sigma_t(r)$ between good and bad times.

As a benchmark, Figure A11 plots the time variation in $\sigma_t(r)$ at the 6-month horizon from the standpoint of an investor with log utility.²² The figure shows that volatility varies from close to 10% in good times to more than 40% in bad times. As such, the price of discount-rate risk must vary significantly and countercyclically to generate the needed variation in ς_t .

Moreover, in the case that $\rho_t(r, \mathbb{E}_{t+1} r) = -1$, then the value $-\varsigma_t$ is a scaled version of $\text{cov}_t(MR_{t,t+1}, r_{t+1})$. Under the *modified negative correlation condition* (mNCC) used throughout Gao and Martin (2021), this covariance is always negative, so ς_t cannot flip sign. This illustrates that the degree of countercyclical variation in discount-rate risk prices required under $\rho_t(r, \mathbb{E}_{t+1} r) = -1$ is quite restrictive, and it is ruled out by a range of standard models considered by Gao and Martin.

In practice, however, we expect $\rho_t(r, \mathbb{E}_{t+1} r)$ to be higher than -1 , as not all changes in stock returns are driven by discount rates. A $\rho_t(r, \mathbb{E}_{t+1} r) > -1$ implies a less volatile price of discount-rate risk than just discussed. However, it also implies that the price of discount-rate risk must be even lower during good times than above; for example, in the opposing extreme in which $\rho_t(r, \mathbb{E}_{t+1} r) = 0$ (i.e., with i.i.d. returns), the Sharpe ratio on the discount-rate claim must be *negative* in good times in order for ς_t to flip sign.

¹⁹As we will show, it is possible for standard models to generate a ς_t that is sufficiently volatile, but if ς_t does not change sign, the average of $\widehat{\varepsilon}$ will be far from zero, inconsistent with the near-zero empirical mean $\widehat{\varepsilon}$.

²⁰That is, $SR_{\mathbb{E},t} = (\mathbb{E}_t[r_{\mathbb{E},t+1}] - r_{t,t+1}^f + \sigma_{\mathbb{E},t}^2/2)/\sigma_{\mathbb{E},t}$, where $r_{\mathbb{E},t+1}$ is the claim’s log return and $\sigma_{\mathbb{E},t}$ is its standard deviation.

²¹This is an extreme case in which all movements in realized returns reflect variation in discount rates (Campbell and Ammer 1993).

²²This is likely to be a conservative benchmark for the degree of variation in conditional volatility if investors are more risk-averse than implied by log utility.

The discussion above can also be generalized beyond the log-normal case. For example, assume that the SDF can be written as

$$M_{t,t+1} = \beta \frac{V_W(W_{t+1}, z_{t+1})}{V_W(W_t, z_t)},$$

where V_W is an unconstrained investor's marginal utility of wealth, z_t is a state vector that includes $-\mathcal{L}_t^{(1)}$, and V_W is weakly decreasing in each entry of z_t . If this agent is fully invested in the market, has relative risk aversion $-WV_{WW}/V_W$ of at least 1, and R_{t+1} and the entries of z_{t+1} are *associated* random variables,²³ then Result 4 of [Gao and Martin \(2021\)](#) can be applied to obtain that ς_t again cannot change sign under the benchmark $\rho_t(r, \mathcal{L}_{t+1}^{(1)}) = -1$.

E.3 Further Intuition Under Power Utility

The above emphasizes that the price of discount-rate risk must vary substantially over time to rationalize our results. To better understand this result, we next re-conduct our analysis under various utility functions generating a constant price of risk, and show how such functions will not be able to explain the data. This analysis will help provide intuition on the role of risk aversion in our results.

We calculate spot and forward rates using CRRA utility with γ ranging from 0.75 to 3. Details are in Appendix F.1. As shown in Table A11, none of the power utility functions can rationalize the results. The functions with $\gamma > 1$ explain the time variation in the forecast errors, but they cannot explain the average being close to zero. To understand the intuition, consider the results derived above for log-normally distributed variables. When $\gamma > 1$, the covariance term ς_t correctly increases in bad times. However, it is always positive ($\text{cov}_t(m_{t+1} + r_{t+1}, \mathbb{E}_{t+1} r_{t+2}) > 0$ at all times), leading to large average forecast errors. The opposite mechanics are at play for $\gamma < 1$, for which $\text{cov}_t(m_{t+1} + r_{t+1}, \mathbb{E}_{t+1} r_{t+2}) < 0$ for all t .

This illustrates the challenge in explaining both the average and time variation in the covariance term, and thus in explaining the results from options through this term alone.

F Additional Empirical Results and Implications

F.1 Power Utility Regressions

This appendix derives the power utility analogue to the LVIX in Appendix E.3. To do so, we apply results from [Breedon and Litzenberger \(1978\)](#) and [Martin \(2017\)](#). We omit time and time-to-maturity dependence throughout for simplicity.

Lemma A1 (Spanning). *For any α and β ,*

$$\frac{1}{R_f} \mathbb{E}_t^* [R^\alpha (\ln R)^\beta] = R_f^{\alpha-1} (\ln R_f)^\beta + \int_0^F \omega(\alpha, \beta) \text{put}(K) dK + \int_F^\infty \omega(\alpha, \beta) \text{call}(K) dK,$$

where

$$\omega(\alpha, \beta) = -\frac{\alpha(1-\alpha)m^\beta + \beta(1-2\alpha)m^{\beta-1} + \beta(1-\beta)m^{\beta-2}}{P_t^2} \left(\frac{K}{P_t}\right)^{\alpha-2} \quad (\text{A14})$$

and $m = \ln K - \ln P_t$.

²³Formally, the elements of the vector $X_{t+1} = (R_{t+1}, z'_{t+1})'$ are associated random variables if $\text{cov}_t(f(X_{t+1}), g(X_{t+1})) \geq 0$ for all nondecreasing functions f and g for which this covariance exists.

As is well-known, under standard regularity conditions, we can compute the price of any function of the index price via a replicating portfolio of bonds, stocks, and options. We simply apply this result to the function $R^\alpha (\ln R)^\beta$, which is useful for power utility expectations below.²⁴

Proposition A1 (Power Utility). *From the standpoint of an unconstrained power utility investor fully invested in the market,*

$$\mathbb{E}_t[\ln R] - \ln R_f = \frac{\mathbb{E}_t^*[R^\gamma \ln R]}{\mathbb{E}_t^*[R^\gamma]} - \ln R_f, \quad (\text{A15})$$

where

$$\frac{1}{R_f} \mathbb{E}_t^*[R^\gamma \ln R] = R_f^{\gamma-1} \ln R_f + \int_0^F \omega(\gamma, 1) \text{put}(K) dK + \int_F^\infty \omega(\gamma, 1) \text{call}(K) dK, \quad (\text{A16})$$

and

$$\frac{1}{R_f} \mathbb{E}_t^*[R^\gamma] = R_f^{\gamma-1} + \int_0^F \omega(\gamma, 0) \text{put}(K) dK + \int_F^\infty \omega(\gamma, 0) \text{call}(K) dK, \quad (\text{A17})$$

and γ is the investor's risk aversion.

The LVIX is a special case of (A15) with $\gamma = 1$, and so the mechanics under power utility are similar, if only messier, to those under log utility. However, there is one caveat: as risk aversion γ increases, the weights on deep out-of-the-money call options in (A14) become untenably large. Since these options are largely unobservable, we can only realistically measure expectations for a $\gamma \leq 3$ investor in practice.

Armed with equity premia from the standpoint of a power utility investor, we can compute spot rates, forward rates, and forecast errors in the usual way. Table A11 reports regression slopes and average errors for different values of risk aversion. Figure A20 visualizes the results. See Appendix E for a more thorough discussion.

F.2 Long-Horizon Forecast Errors

This appendix provides details on the forecast-error quantification in Section 5.1.

For the quantification, we re-estimate spot rates, forward rates, and forecast errors at longer horizons (up to $m + n = 8$ years) for the Euro Stoxx 50 (SX5E). The sample runs from September 2005 through September 2014 (beyond which we cannot yet observe realized forecast errors). The combinations of m and n (in months) can be seen in Table A12.

We are only able to do this analysis for the Euro Stoxx 50. No other exchange has comparably long-maturity options available for such a long period. The only other exchange with such long maturities is the FTSE 100, but this sample only starts in 2014. More importantly, long-maturity options are liquid for the SX5E: the share of contracts with positive open interest is very consistent for medium (1 to 3 years) and long (3 to 10 years) maturities. In contrast, for the FTSE 100, there is a severe decline in the relative liquidity of long-maturity options (that is, almost no liquidity as per open interest). A few exchanges in the main sample do have options with maturities up to 5 years, but again, the liquidity of these options is suspect, rendering them unusable, at least for this analysis.

²⁴Lemma A1 clarifies the links among several volatility indexes. The VIX index from the CBOE is a function of $\mathbb{E}_t^*[\ln R]$, the SVIX index from Martin (2017) a function of $\mathbb{E}_t^*[R^2]$, and the LVIX index from Gao and Martin (2021) a function of $\mathbb{E}_t^*[R \ln R]$.

For each such combination of m and n , we predict forecast errors as in (9) using a regression of realized forecast errors on shorter-horizon forward rates; we use the $(n - 12) \times 12$ forward rate (with horizons again now in months) for $n \geq 24$, and for $n = 12$ we use the 6×6 rate. After obtaining these predicted forecast errors, we calculate a decay parameter for each date’s forecast errors, $\phi_t^{(n,m)}$, as the ratio of estimated $\mathbb{E}_t[\tilde{\varepsilon}_{t+n}^{(m)}]$ to $\mathbb{E}_t[\tilde{\varepsilon}_{t+12}^{(m)}]$ for each available $m, n > 12$. This decay specification builds on the one used by [De la O and Myers \(2021, eq. \(13\)\)](#) for expected returns, but we estimate it directly for each date t (whereas they use a full-sample regression for one horizon).

The entries of Table [A12](#) report the median decay parameter over all t for each combination of m and n . In all cases the estimates are close to or above 1. Assuming that predictable forecast errors are permanent at all horizons might be thought of as providing an estimate of their maximal possible effect. That said, when we estimate the decay parameter in the U.S. (at shorter horizons, unreported), we in fact generally obtain estimates greater than 1, suggesting that setting $\phi^{(n,m)} = 1$ may, if anything, be slightly conservative in the U.S. sample.

F.3 Impact of Forecast Errors on Demand Elasticities

In addition to the implications discussed in Section 5 of the main paper, our results may also help speak to the recent finding that investors’ demand for equities is relatively inelastic with respect to price changes ([Gabaix and Koijen 2022](#)). Empirically, there is a strong relation between changes in prices and expected one-period returns. [Nagel and Xu \(2023\)](#), for instance, show that a 1 standard deviation increase in the dividend yield increases expected stock returns by 6 percentage points over the subsequent year. This large impact on expected returns should in principle lead investors to allocate substantially towards equities when prices move, making their documented demand inelasticity appear puzzling. But our results suggest that investors often mistakenly attribute a significant share of the decrease in prices to increases in risk premia at relatively long horizons, and perceive short-term spot risk premia to increase only modestly. A modest increase in short-term risk premia will not lead to a large increase in desired portfolio weights in equities, providing a candidate explanation — albeit a speculative one — for the observed inelasticity. We leave potential quantification of this effect to future work.

G Expectations-Based Explanation for Empirical Results

G.1 A Model of Expectation Errors

To provide a more positive potential explanation for our results, this appendix presents a simple model of expectation errors in the term structure of spot rates. The model builds on the fundamental extrapolation model proposed by [Hirshleifer, Li, and Yu \(2015\)](#), but it is also qualitatively similar to the natural expectations model proposed by [Fuster, Laibson, and Mendel \(2010\)](#) and adapted by [Giglio and Kelly \(2018\)](#). The model matches both the excess cyclicalities in forward rates across all investors and the cyclicalities of short-term spot rates across different investors.

G.1.1 Model Setup

We first describe the rational benchmark. The starting point is a state variable x_t . In our context, this fundamental state variable is most straightforwardly interpreted as representing market-level variance, but this labeling is not necessary for what follows. Under the objective measure, the

fundamental variable follows an AR(3) process:

$$x_t = \left(1 - \sum_{j=1}^3 \phi_j\right) \bar{x} + \phi_1 x_{t-1} + \phi_2 x_{t-2} + \phi_3 x_{t-3} + e_t, \quad (\text{A18})$$

where $\bar{x} = \mathbb{E}[x_t]$ and $e_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_e^2)$. We assume that x_t is countercyclical. This is without loss of generality, though it applies naturally to variance. Under a rational benchmark, investors use the objective dynamics to make iterated forecasts of fundamentals:

$$\mathbb{E}_t[x_{t+n}] = \left(1 - \sum_{j=1}^3 \phi_j\right) \bar{x} + \sum_{j=1}^3 \phi_j \mathbb{E}_t[x_{t+n-j}] \text{ for } n > 0. \quad (\text{A19})$$

We next turn to equity premia. There is an affine mapping between contemporaneous fundamentals and short-term spot rates (henceforth *short rates* for short). Under rationality, the current short rate is:²⁵

$$\mathbb{E}_t[\mu_t^{(3)}] = \alpha_0 + \alpha_1 x_t. \quad (\text{A20})$$

We assume that $\alpha_1 > 0$, so that the objective equity premium is countercyclical, as is standard. Investors use this mapping to translate from forecasts of fundamentals to forecasts of future short rates:

$$\mathbb{E}_t[\mu_{t+n-3}^{(3)}] = \alpha_0 + \alpha_1 \mathbb{E}_t[x_{t+n-3}] \text{ for } n \in \{6, 9, 12\}. \quad (\text{A21})$$

Under rationality, objective short rate expectations define long-term spot rates, as in (2):

$$\mathbb{E}_t[\mu_t^{(n)}] = \mathbb{E}_t[\mu_t^{(n-3)}] + \mathbb{E}_t[\mu_{t+n-3}^{(3)}] \text{ for } n \in \{6, 9, 12\}. \quad (\text{A22})$$

With this term structure of spot rates in hand, we can compute forward rates and forecast errors in the usual way.

We now turn to our assumptions on investors' actual subjective expectations. Investors observe current fundamentals x_t , and to limit free parameters, we also assume that they know the persistence parameters ϕ_1, ϕ_2 , and ϕ_3 in (A18) perfectly. We assume two deviations from rationality. The first affects how investors forecast future short rates. We refer to this as *forward rate bias* because it distorts long-term spot rates via forward rates. The second affects how investors perceive current short rates. We refer to this as *short rate bias* because it distorts long-term spot rates via current short rates. In what follows, we use $\mathbb{E}_t^\theta[\cdot]$ to differentiate subjective from objective expectations: θ_F indexes forward rate bias, θ_S short rate bias. We discuss each in turn.

We begin with forward rate bias. Investors forecast future fundamentals using (A19), but to match the excess cyclicity in forward rates, we assume that they do so with a misperceived long-term mean:

$$\mathbb{E}_t^\theta[\bar{x}] = \mathbb{E}_{t-1}^\theta[\bar{x}] + \underbrace{\theta_F \left(x_t - \mathbb{E}_{t-1}^\theta[\bar{x}]\right)}_{\text{overreaction to news}}. \quad (\text{A23})$$

While the objective long-run mean $\bar{x}$ is constant, investors do not treat it as such, and they attach

²⁵We abuse notation slightly in denoting $\mu_t^{(3)}$ as $\mathbb{E}_t[\mu_t^{(3)}]$, which is somewhat redundant given that $\mu_t^{(3)} = \mathbb{E}_t[r_{t,t+3}]$. We do so to emphasize that the spot rate is itself an expectation (while maintaining focus on spot rates). This will also allow for a clearer distinction vis-à-vis subjective expectations below.

too much weight to recent fundamentals in updating about $\bar{x}$. This specification of extrapolative expectations is from [Hirshleifer, Li, and Yu \(2015\)](#), and we apply it to the long-term mean in the spirit of [Nagel and Xu \(2022\)](#) and [Afrouzi et al. \(2023\)](#), among others.²⁶ This representation nests rationality when $\theta_F = 0$. When $\theta_F > 0$, subjective expectations of the long-term mean overreact to perceived news $\hat{e}_t = x_t - \mathbb{E}_{t-1}^\theta[\bar{x}]$. In bad times, for example, overreaction leads to higher forecasts of fundamentals via (A19), and thus higher forecasts of short rates via (A21) and higher forward rates via (A22), replacing objective with subjective expectations $\mathbb{E}_t^\theta[\cdot]$. This generates predictably negative forecast errors, thereby allowing the model to match the excess cyclicity in forward rates.

We next turn to short rate bias. To match the cyclicity of short rates, we assume that investors incorrectly map from fundamentals to current short rates as follows:

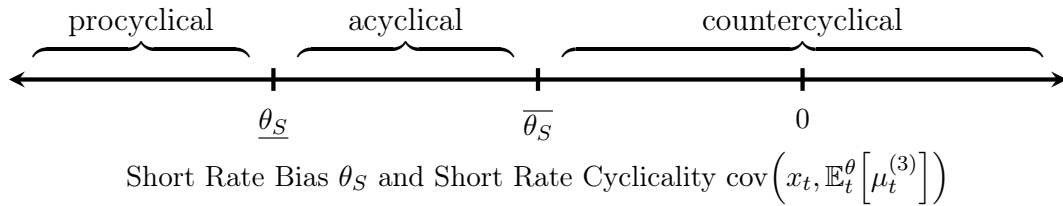
$$\mathbb{E}_t^\theta[\mu_t^{(3)}] = \mathbb{E}_t[\mu_t^{(3)}] + \alpha_1 \theta_S e_t = \alpha_0 + \alpha_1 (x_t + \theta_S e_t). \quad (\text{A24})$$

While investors understand that short rates increase in the fundamental state variable, they also include the most recent shock in this mapping.²⁷ This representation nests rationality when $\theta_S = 0$. When $\theta_S \neq 0$, there is a wedge between objective and subjective short rates, as shown on the left. This wedge is proportional to the most recent shock to fundamentals. Equivalently, as shown on the right, investors behave as if they misperceive fundamentals. This interpretation does not hold literally (we assume that investors observe fundamentals correctly), but it may be more intuitive.

The covariance between fundamentals and short rates, from the standpoint of the econometrician, measures the cyclicity of short rates:

$$\underbrace{\text{cov}\left(x_t, \mathbb{E}_t^\theta[\mu_t^{(3)}]\right)}_{\text{subjective}} = \underbrace{\text{cov}\left(x_t, \mathbb{E}_t[\mu_t^{(3)}]\right)}_{\text{objective}} + \underbrace{\alpha_1 \theta_S \text{cov}(x_t, e_t)}_{\text{wedge}}.$$

As above, the covariance between fundamentals and subjective short rates (the subjective covariance) equals the corresponding covariance with objective short rates (the objective covariance) plus a wedge. The objective covariance is positive, but the wedge (and thus the subjective covariance) can be positive or negative. The schematic diagram below visualizes the relationship between short rate bias and the cyclicity of short rates:



For sufficiently small $\theta_S < \underline{\theta_S}$, the subjective covariance is negative and so short rates are procyclical

²⁶[Hirshleifer, Li, and Yu \(2015\)](#) assume that the one-period forecast for a fundamental growth rate $\hat{x}_{t,t+1}$ follows $\hat{x}_{t,t+1} = (1 - \rho - \tilde{\rho})\bar{x} + \rho\hat{x}_{t-1,t} + \tilde{\rho}\hat{e}_t$, where $\hat{e}_t$ is perceived news. They typically assume the sum $\rho + \tilde{\rho}$ is close to but just less than 1, to ensure finiteness of the value function in their general equilibrium model. We do not face this issue, so we set $\rho + \tilde{\rho} = 1$, which — setting $\tilde{\rho} = \theta_F$ — generates (A23). We note that time variation in the perceived long-run mean need not be a mistake in general: investors may be learning about this parameter. But the constant-gain learning implied by (A23) implies that this learning is overattentive to recent news and never converges. For further intuition, following [Hirshleifer, Li, and Yu, \(A23\)](#) can be rewritten as $\mathbb{E}_t^\theta[\bar{x}] = (1 - \theta_F)\mathbb{E}_{t-1}^\theta[\bar{x}] + \theta_F x_t$. In this formulation, the first term reflects the persistence of extrapolation from past fundamentals, the second term extrapolation from current fundamentals.

²⁷For parsimony, we set this term to e_t so that we have only a single shock.

(Greenwood and Shleifer 2014). For $\underline{\theta}_S < \theta_S < \overline{\theta}_S$, it is approximately zero and so short rates are acyclical (Nagel and Xu 2023). For sufficiently large $\theta_S > \overline{\theta}_S$, it is positive and so short rates are countercyclical (Dahlquist and Ibert 2024). In this way, the model can match the cyclicity of short rates for multiple groups of investors.

We think of short rate bias as a stand-in for different investors' (mis)reactions to short-term data. For $\theta_S < \underline{\theta}_S < 0$, the bias can be thought of as a failure to learn from prices (Eyster, Rabin, and Vayanos 2019): after a positive shock to fundamentals (volatility) e_t , it is as if investors believe only they have observed the shock and that prices will continue to fall when others observe the shock (generating extrapolation). By contrast, $\theta_S > 0$ can be thought of as representing excessive learning from prices (Bastianello and Fontanier 2025), generating excessively high beliefs about future volatility and expected returns.

Finally, we note in passing that the short rate wedge $\alpha_1 \theta_S e_t$ does not affect forecasts of future short rates: by assumption, $\mathbb{E}_t^\theta[e_{t+n}] = \mathbb{E}_t[e_{t+n}] = 0$ for $n > 0$. As a result, forward rate bias is separate from short rate bias.

To summarize, our two assumed biases have distinct effects on the empirical results. Forward rate bias distorts long-term spot rates via forward rates. It affects Mincer-Zarnowitz and error-predictability regression slopes, but short rate bias does not. Short rate bias distorts long-term spot rates via short rates. It affects the cyclicity of short rates, but forward rate bias does not. This disconnect between the two biases (and thus the corresponding rates) is important. It allows the model to match both the excess cyclicity in forward rates across all investors and the cyclicity of short rates across different investors; that is, the *same* forward-rate dynamics for *different* short-rate dynamics.²⁸

G.1.2 Calibration and Results

The discussion above suggests the model can qualitatively match the excess cyclicity in forward rates. To understand the quantitative implications, we turn to Monte Carlo simulations. To calibrate the model, we assume that the fundamental variable x_t is equal to option-implied variance (in the spirit of Figure A3, and as discussed above) and estimate its dynamics in the U.S. sample. We measure spot rates using our U.S. option-implied risk premia and estimate the mapping from fundamentals to spot rates. We then consider a range of θ_F and θ_S parameters. For each combination, we solve for the 6-month, 6-month forward rate, simulating 100,000 samples of length $T = 378$ months. In the simulated data, we evaluate how regression slopes and average errors vary with deviations from rationality. We compare the simulated moments with those from the option-based measure and the Livingston survey. Details of the calibration and simulations are in Appendix G.2.1 below.

Panel A of Figure A12 plots the main results from these simulations, showing regression slopes and average errors. Since short rate bias affects only regression R^2 values and not slopes or average errors, we set $\theta_S = 0$ for these figures and focus on how the results depend on θ_F . Overall, we find that a model with forward rate bias is consistent with the data. Starting with $\theta_F = 0$, the model recovers the rational null for both slopes and average errors. But as θ_F increases, the Mincer-Zarnowitz slope decreases toward zero and the error-predictability slope becomes more negative. Given (A23), subjective expectations are on average unbiased, so the model cannot produce any non-zero average error. There is thus no relationship between θ_F and the average error. For the option-based measure, $\theta_F = 0.0516$ exactly matches the slopes and closely matches the average

²⁸Both biases affect Mincer-Zarnowitz and error-predictability R^2 values, but neither bias affects average errors because $\mathbb{E}[\mathbb{E}_t^\theta[\bar{x}]] = \bar{x}$ in (A23) and $\mathbb{E}[e_t] = 0$ in (A24).

error. This value is slightly higher than typical estimates in the literature, but those estimates are from very different settings than the one considered here; see [Hirshleifer, Li, and Yu \(2015\)](#) for a discussion. For the Livingston survey, the same value fits the slopes reasonably well, but misses the average error. In any case, fundamental extrapolation seems to explain the excess cyclical in forward rates well for all investors.

G.1.3 Interpretation and Discussion

While the above model of expectation errors is capable of matching our empirical findings, the cyclical in expectation errors distinguishes our setting from certain frameworks featuring return extrapolation. In the model, forward rates overreact to fundamental news: when the objective equity premium is high, forward rates increase, and forecast errors are predictably negative. This echoes our empirical findings. Those findings would not be matched by specifying a model of overreaction of forward rates to realized returns: the estimated equity premium is high in bad times, when realized returns are *low*.

If the cyclical in our model’s expectation errors were to be flipped — if forward rates were too low in bad times, generating positive errors — then these errors would not be capable of explaining any of the variation in the price-dividend ratio shown in [Figure A4](#). Forecast errors instead would exert a *dampening* force on stock prices, making it more difficult to explain their observed volatility. We discuss and formalize this idea in more detail in [Appendix G.2.2](#), which introduces a “trilemma” for expectation errors: it is difficult to simultaneously make sense of (i) volatile expectation errors, (ii) countercyclical expectation errors (e.g., extrapolation of past returns), and (iii) volatile prices. Our framework discards (ii), leaving (i) and (iii) on the table.

G.2 Further Model Details

G.2.1 Calibration Implementation

Fundamentals. We measure fundamentals with option-implied volatility (variance, not standard deviation). We omit time and horizon dependence below for simplicity. From the standpoint of an unconstrained log utility investor fully invested in the market, the conditional variance of the 3-month market return is:

$$x_t \equiv \sigma_t^2(\ln R) = \mathbb{E}_t[(\ln R)^2] - (\mathbb{E}_t[\ln R])^2 = \frac{\mathbb{E}_t^*[R(\ln R)^2]}{\mathbb{E}_t^*[R]} - \left(\frac{\mathbb{E}_t^*[R \ln R]}{\mathbb{E}_t^*[R]} \right)^2.$$

As before, we consider a log utility investor ([Martin 2017](#)) and compute risk-neutral expectations via [Lemma A1](#). The objective volatility dynamics in ([A18](#)) are

ϕ_1	ϕ_2	ϕ_3	$\bar{x}$	σ_e
0.96	-0.27	0.17	0.98	0.36
(0.05)	(0.07)	(0.05)	(0.13)	(0.01)

The units are non-annualized percentage points. Standard errors are from the simulations below.²⁹

Simulations. We evaluate the model via Monte Carlo simulations. The steps are as follows:

²⁹We note that any measure of volatility would produce qualitatively similar results: what matters is not the particular measure but rather the mapping to risk premia.

1. We simulate the dynamics of volatility via (A18). We simulate 100,000 artificial samples of length $T = 378$ months. In each artificial sample, we use a burn-in period of 100 years.
2. We compute forecasts of future volatility via (A19), but we replace the objective long-term mean $\bar{x}$ with the misperceived counterpart in (A23). In each artificial sample, we initialize the misperceived mean at $\bar{x}$. When $\theta_F > 0$, these forecasts are distorted by forward rate bias; $\theta_F = 0$ nests rationality.
3. We map forecasts of future volatility to forecasts of future short rates via (A21), but we replace the objective expectation $\mathbb{E}[\cdot]$ with the subjective counterpart $\mathbb{E}^\theta[\cdot]$ from the previous step. We estimate the mapping in (A21) from a regression of contemporaneous short rates on volatility, as in (A20).
4. We compute perceptions of current short rates via (A24). When $\theta_S \neq 0$, these perceptions are distorted by short rate bias; $\theta_S = 0$ nests rationality.
5. We compute current long-term spot rates via (A22), but we again replace the objective expectation $\mathbb{E}[\cdot]$ with the subjective counterpart $\mathbb{E}^\theta[\cdot]$ from the previous two steps.

With the term structure of spot rates in hand, we can compute forward rates and forecast errors in the usual way.

G.2.2 Discussion: A Trilemma for Expectation Errors

This appendix continues the discussion in Appendix G.1.3 on how different moments of the data are tied together by the cyclicalities of forecast errors.³⁰ We begin with the Campbell-Shiller price-dividend decomposition in (10). Assume that the expectations $\mathbb{E}_t[\cdot]$ in that decomposition refer to agents' subjective beliefs, and $p_t - d_t$ is the observed log price-dividend ratio. Now consider an alternative economy in which all agents have rational expectations. For arbitrary equilibrium variable x_t in the observed data, denote the corresponding variable in the alternative RE economy by x_t^{RE} . Define the wedge between these two variables to be $\widetilde{x}_t = x_t - x_t^{RE}$. For example, $\widetilde{p_t - d_t}$ is the wedge between the observed price-dividend ratio and the one that would be observed in the alternative economy with RE. Up to a constant, it satisfies

$$\widetilde{p_t - d_t} = \widetilde{CF_t} - \widetilde{\mathcal{F}_t} - \widetilde{RF_t}. \quad (\text{A25})$$

Assume for simplicity that $\widetilde{RF_t} = 0$. The following variance decomposition for the price-dividend wedge therefore holds:

$$\text{var}(\widetilde{p_t - d_t}) = \text{var}(\widetilde{CF_t}) + \text{var}(\widetilde{\mathcal{F}_t}) - 2 \text{cov}(\widetilde{CF_t}, \widetilde{\mathcal{F}_t}). \quad (\text{A26})$$

Alternatively, one can also use the following decomposition given (A25):

$$\text{var}(\widetilde{p_t - d_t}) = \text{cov}(\widetilde{p_t - d_t}, \widetilde{CF_t}) - \text{cov}(\widetilde{p_t - d_t}, \widetilde{\mathcal{F}_t}). \quad (\text{A27})$$

The wedges $\widetilde{CF_t}$ and $\widetilde{\mathcal{F}_t}$ can be understood as expectation errors along the lines considered in Appendix G.1: if subjective expectations are too high relative to RE, then the wedge will be positive

³⁰This discussion builds on Campbell (2017), who introduces related trilemmas for present value and portfolio choice.

(and forecast errors, defined as realized – forecast, are likely to be negative). According to either of the decompositions in (A26)–(A27), therefore, one must choose from at most two of the following three features of any model of expectation errors:

1. Volatile expectation errors for returns (and/or fundamentals)
2. Volatile price-dividend ratio relative to a rational benchmark
3. Countercyclical return expectation errors (positive return expectation errors in bad times)

For example, if excessively positive cash-flow and return forecast revisions occur in good times (after positive news), then $\text{cov}(\widetilde{CF}_t, \widetilde{\mathcal{F}}_t) > 0$ in (A26). Alternatively, in the version expressed in (A27), positive comovement between price-dividend and forward-rate wedges similarly detracts from a model’s ability to generate volatile $\widetilde{p_t - d_t}$. This form of overreaction to *realized outcomes* (cash flows and/or returns) may be intuitively appealing, but it limits a model’s ability to speak to variation in the price-dividend ratio through expectation errors alone.³¹

Our empirical results, and our model of expectation errors, instead suggest overreaction of forward rates to fundamentals, rather than realized returns. Unlike realized returns, we find that spot and forward rates *increase* in bad times. The negative covariance between fundamental news and return expectation errors in principle allows for a volatile price-dividend ratio.

³¹For example, Nagel and Xu (2022) obtain a price-dividend ratio volatility about 50% lower than that observed in the data (see their Table 5). Similarly, De la O and Myers (2021) report that in the model of Barberis et al. (2015), “movements in dividend change expectations are almost completely negated by movements in price change expectations. This leads to low variation in the price-dividend difference” (p. 1370); Campbell (2017) provides a related discussion of the Barberis et al. (2015) results.

Appendix Tables and Figures

Table A1
Option Sample

This table reports the country or index name, the abbreviation, the underlying index, the sample period, and the sample length in months for each exchange. The last column indicates whether the exchange is in the main sample. See Appendix C.1 for more details.

Name	Abbrv	Index	Start	End	Length	Main
North America						
United States	USA	S&P 500	199001	202112	384	Y
Europe						
Belgium	BEL	BEL 20	200201	202109	228	
Switzerland	CHE	SMI	200201	202109	237	Y
Germany	DEU	DAX	200201	202109	237	Y
Spain	ESP	IBEX 35	200610	202109	180	Y
Finland	FIN	OMXH25	200201	202109	237	
France	FRA	CAC 40	200304	202109	222	Y
United Kingdom	GBR	FTSE 100	200201	202109	237	Y
Italy	ITA	FTSE MIB	200610	202109	180	Y
Netherlands	NLD	AEX	200201	202109	219	
Sweden	SWE	OMXS30	200705	202109	173	
Pan-Europe						
Euro Stoxx 50	SX5E	SX5E	200201	202109	237	Y
Stoxx Europe 50	SX5P	SX5P	200201	202109	237	
Stoxx Europe 600	SXXP	SXXP	200509	202109	193	
Asia-Pacific						
Australia	AUS	ASX 200	200401	202104	208	Y
China	CHN	HSCEI	200601	202104	184	
Hong Kong	HKG	Hang Seng	200601	202104	184	Y
Japan	JPN	Nikkei 225	200405	202104	204	
Korea	KOR	KOSPI 200	200407	202104	202	
Taiwan	TWN	TAIEX	200510	202104	187	

Table A2
Summary Statistics

This table reports summary statistics for option-based (Panel A) and survey-based (Panel B) expectations. The expectations are forward rates and the corresponding realized spot rates for risk premia (on the left) and expected returns (on the right). The option-based horizon is the 6×6 -month forecast error, the Livingston horizon is the 6×6 -month forecast error, the CFO horizon is the 1×9 -year forecast error, and the Vanguard horizon is the 1×9 -year forecast error. The units are annualized percentage points. The option-based sample is the longest available for each exchange. The Livingston sample is half-yearly from 06/1992 to 06/2021 for forward rates and from 12/1992 to 12/2021 for realized spot rates. The CFO sample is quarterly from 12/2001 to 12/2018 for forward rates and from 12/2002 to 12/2019 for realized spot rates. The Vanguard sample is bimonthly from 02/2017 to 06/2021 (including 03/2020). See Section 4 for more details.

PANEL A. OPTION-BASED EXPECTATIONS									
	Forward RP		Spot RP		Forward ER		Spot ER		
	Mean	StDev	Mean	StDev	Mean	StDev	Mean	StDev	Length
North America									
USA	2.15	0.99	2.17	1.40	5.44	2.32	5.04	2.41	378
Europe									
BEL	2.56	2.22	2.42	2.07	4.13	2.79	3.59	2.82	213
CHE	1.88	0.80	1.96	1.34	2.60	1.50	2.26	1.85	231
DEU	2.63	1.07	2.85	1.86	4.27	2.12	4.10	2.74	231
ESP	2.89	1.17	3.37	1.83	4.19	2.19	4.19	2.73	168
FIN	2.38	1.85	2.79	2.21	4.02	2.65	4.03	2.85	231
FRA	2.44	1.04	2.57	1.56	3.97	2.07	3.71	2.42	215
GBR	2.12	1.00	2.21	1.51	4.76	2.13	4.33	2.59	224
ITA	3.17	1.35	3.60	1.67	4.42	2.23	4.39	2.40	174
NLD	2.43	1.26	2.85	2.18	3.95	2.40	3.93	3.20	202
SWE	2.44	1.41	2.79	1.80	3.77	2.68	3.79	2.83	160
Pan-Europe									
SX5E	2.65	1.08	2.91	1.83	4.29	2.10	4.15	2.70	231
SX5P	2.11	1.48	2.23	1.62	3.75	2.51	3.47	2.64	231
SXXP	2.05	1.53	2.14	1.63	3.44	2.57	3.12	2.65	185
Asia-Pacific									
AUS	1.84	1.20	1.97	1.39	5.95	2.39	5.65	2.65	197
CHN	3.90	3.29	4.24	4.01	5.92	3.64	5.61	4.42	178
HKG	2.85	1.93	2.98	2.56	4.86	2.26	4.35	2.98	178
JPN	2.60	1.57	2.93	2.20	3.15	1.76	3.21	2.35	197
KOR	2.14	1.85	2.23	2.16	5.40	2.95	5.13	3.02	194
TWN	2.47	1.53	2.39	1.90	3.95	1.74	3.37	2.11	181

(Continued on the next page)

Table A2
Summary Statistics (Continued)

PANEL B. SURVEY-BASED EXPECTATIONS									
	Forward RP		Spot RP		Forward ER		Spot ER		Length
	Mean	StDev	Mean	StDev	Mean	StDev	Mean	StDev	
Livingston Survey									
Option	2.16	1.01	2.08	1.20	5.20	2.16	4.79	2.24	59
Survey	3.15	4.96	3.52	5.35	5.73	3.81	5.84	4.14	59
CFO Survey									
Option	2.29	1.08	2.21	1.50	4.22	1.45	3.95	1.92	68
Survey	3.37	0.57	3.46	0.57	6.77	0.77	6.52	0.68	68
Vanguard Survey									
Option	1.98	0.63	2.25	1.18	3.48	0.60	3.60	0.92	27
Survey	4.61	0.95	4.67	0.95	6.67	0.28	6.58	0.30	27

(Continued from the previous page)

Table A3
Option-Based Expectations: Instrumented Mincer-Zarnowitz Regressions

This table reports instrumented [Mincer-Zarnowitz](#) regressions of future realized spot rates on current forward rates for option-based risk premia (left panel) and expected returns (right panel). For risk premia, the realized spot rate is the future expectation of the 6-month equity premium, and the forward rate is the current expectation of the same risk premium. The instrument is the current expectation of the 1-month equity premium in 2 months:

$$\tilde{f}_t^{(2,1)} = \mathbb{E}_t \left[r_{t+2}^{(1)} \right].$$

These expectations are analogously defined for expected returns. The units are annualized percentage points. Panel regressions, in the main sample, report standard errors clustered by exchange and date. This sample is the longest available for each exchange. Time-series regressions, in the U.S. sample, report [Newey-West](#) standard errors with $L = \lceil 1.3 \times T^{1/2} \rceil$ lags and fixed- b p -values, following [Lazarus et al. \(2018\)](#). This sample is from 01/1990 to 06/2021 for forward rates and from 07/1990 to 12/2021 for realized spot rates. See Section 4.1 for more details.

	RISK PREMIA $\tilde{\mu}_{t+6}^{(6)}$			EXPECTED RETURNS $\mu_{t+6}^{(6)}$		
	(1) Main	(2) Main	(3) U.S. Only	(4) Main	(5) Main	(6) U.S. Only
$\tilde{f}_t^{(6,6)}$	0.77 (0.071)	0.70 (0.074)	0.73 (0.10)			
$f_t^{(6,6)}$				0.93 (0.060)	0.93 (0.072)	0.89 (0.081)
Intercept	0.73 (0.19)		0.60 (0.31)	0.024 (0.24)		0.21 (0.39)
p -val: $\beta_1 = 1$	0.001	0.003	0.018	0.274	0.328	0.182
p -val: $\beta_0 = 0$	<0.001	-	0.065	0.921	-	0.590
p -val: $\beta_1 = 1, \beta_0 = 0$	0.001	-	0.058	0.018	-	0.111
Observations	2227	2227	378	2227	2227	378
Fixed Effects	None	Ex	None	None	Ex	None
Standard Errors	Cluster	Cluster	Newey-West	Cluster	Cluster	Newey-West
Clusters/Lags	Ex/Date	Ex/Date	26	Ex/Date	Ex/Date	26
Adjusted R^2	0.19	0.21	0.22	0.63	0.63	0.71
Within R^2	-	0.14	-	-	0.59	-

Table A4
Predictability of Survey-Based Forecast Errors with Option-Based Forward Rates

This table reports predictability regressions of future realized forecast errors on current forward rates for risk premia (left panel) and expected returns (right panel). On the left-hand side, the option-based horizon is the 6×6 -month forecast error, the Livingston horizon is the 6×6 -month forecast error, and the CFO horizon is the 1×9 -year forecast error. On the right-hand side, the predictor is the option-based 2×1 -month forward risk premium. Forecast errors and forward rates are standardized to unit variance. The intercept is not reported. Each regression reports [Newey-West](#) standard errors with $L = \lceil 1.3 \times T^{1/2} \rceil$ lags and fixed- b p -values, following [Lazarus et al. \(2018\)](#). The Livingston sample is half-yearly from 06/1992 to 06/2021 for forward rates and from 12/1992 to 12/2021 for realized spot rates. The CFO sample is quarterly from 12/2001 to 06/2021 (excluding 09/2019 and 12/2019) for forward rates and from 12/2002 to 06/2022 (excluding 09/2020 and 12/2020) for realized spot rates. FedInd is an indicator variable that equals 1 on and after 09/2020 for forward rates and likewise 09/2021 for realized spot rates. See Section 4.3 for more details.

	RISK PREMIA				EXPECTED RETURNS			
	Livingston Survey		CFO Survey		Livingston Survey		CFO Survey	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Option $\tilde{\varepsilon}_{t+6}^{(6)}$	Survey $\tilde{\varepsilon}_{t+6}^{(6)}$	Option $\tilde{\varepsilon}_{t+6}^{(6)}$	Survey $\tilde{\varepsilon}_{t+1y}^{(9y)}$	Option $\varepsilon_{t+6}^{(6)}$	Survey $\varepsilon_{t+6}^{(6)}$	Option $\varepsilon_{t+6}^{(6)}$	Survey $\varepsilon_{t+1y}^{(9y)}$
Option $\tilde{f}_t^{(2,1)}$	-0.32 (0.078)	-0.22 (0.078)	-0.29 (0.10)	-0.28 (0.15)	-0.36 (0.13)	-0.24 (0.078)	-0.47 (0.092)	-0.42 (0.090)
p -val: $\beta_1 = 0$	0.004	0.033	0.027	0.136	0.028	0.019	0.001	0.001
Observations	59	59	72	72	59	59	72	72
Controls	None	None	FedInd	FedInd	None	None	FedInd	FedInd
Newey-West Lags	10	10	12	12	10	10	12	12
Adjusted R^2	0.09	0.03	0.06	0.08	0.12	0.04	0.19	0.21

Table A5
Alternative Integration Bounds

This table reports estimates with alternative integration bounds for option-based risk premia (Panel A) and expected returns (Panel B). [Mincer-Zarnowitz](#) regressions test $H_0: \beta_1 = 1$, as in Table 2. The average forecast error tests $H_0: \bar{\varepsilon}_t = 0$, as in Table 3. Error-predictability regressions test $H_0: \beta_1 = 0$, as in Table 4. The horizon is the 6-month spot rate, 6 months from now. The units are annualized percentage points. All regressions include exchange fixed effects and compute a within R^2 . Standard errors are clustered by exchange and date. The sample is the longest available for each exchange in the main sample. See Appendix [D.1](#) for more details.

PANEL A. RISK PREMIA $\tilde{\mu}_{t+6}^{(6)}$												
	Mincer-Zarnowitz				Average Error			Error Predictability				N
	β_1	$se(\beta_1)$	p -val	R^2	$\tilde{\varepsilon}_t$	$se(\tilde{\varepsilon}_t)$	p -val	β_1	$se(\beta_1)$	p -val	R^2	
Without Extrapolation												
Truncation: $0.75 \leq K/P \leq 1.25$	0.92	0.12		0.19	0.44	0.088	***	0.067	0.040		0.01	2140
Truncation: $0.65 \leq K/P \leq 1.35$	0.78	0.091	**	0.18	0.34	0.094	***	-0.019	0.039		0.00	2139
Truncation: $0.55 \leq K/P \leq 1.45$	0.68	0.074	***	0.17	0.26	0.098	**	-0.082	0.041	*	0.01	2137
Truncation: $0.45 \leq K/P \leq 1.55$	0.63	0.065	***	0.16	0.22	0.099	*	-0.12	0.042	**	0.02	2140
Truncation: $0.35 \leq K/P \leq 1.65$	0.59	0.059	***	0.15	0.20	0.10	*	-0.15	0.044	***	0.03	2138
Observable Strike Prices	0.56	0.053	***	0.15	0.19	0.10	*	-0.17	0.043	***	0.04	2140
With Extrapolation												
Static Integration Bounds	0.56	0.056	***	0.15	0.14	0.10		-0.15	0.041	***	0.04	2244
Dynamic Integration Bounds	0.56	0.055	***	0.15	0.16	0.11		-0.17	0.046	***	0.03	2241
Baseline: $0.10 \leq K/P \leq 2.00$	0.56	0.055	***	0.15	0.17	0.11		-0.16	0.047	***	0.03	2227

PANEL B. EXPECTED RETURNS $\mu_{t+6}^{(6)}$												
	Mincer-Zarnowitz				Average Error			Error Predictability				N
	β_1	$se(\beta_1)$	p -val	R^2	ε_t	$se(\varepsilon_t)$	p -val	β_1	$se(\beta_1)$	p -val	R^2	
Without Extrapolation												
Truncation: $0.75 \leq K/P \leq 1.25$	1.03	0.052		0.77	-0.0070	0.089		-0.12	0.048	**	0.02	2140
Truncation: $0.65 \leq K/P \leq 1.35$	1.02	0.053		0.71	-0.11	0.094		-0.18	0.043	***	0.04	2139
Truncation: $0.55 \leq K/P \leq 1.45$	0.99	0.054		0.67	-0.19	0.098	*	-0.23	0.044	***	0.07	2137
Truncation: $0.45 \leq K/P \leq 1.55$	0.96	0.055		0.64	-0.23	0.100	**	-0.26	0.045	***	0.09	2140
Truncation: $0.35 \leq K/P \leq 1.65$	0.94	0.056		0.63	-0.25	0.10	**	-0.29	0.046	***	0.10	2138
Observable Strike Prices	0.93	0.058		0.61	-0.26	0.10	**	-0.31	0.046	***	0.12	2140
With Extrapolation												
Static Integration Bounds	0.92	0.060		0.60	-0.31	0.10	**	-0.26	0.039	***	0.12	2244
Dynamic Integration Bounds	0.91	0.061		0.59	-0.28	0.10	**	-0.30	0.047	***	0.11	2241
Baseline: $0.10 \leq K/P \leq 2.00$	0.91	0.061		0.59	-0.27	0.10	**	-0.29	0.048	***	0.10	2227

Table A6
Alternative Measures

This table reports estimates with alternative measures for option-based risk premia (Panel A) and expected returns (Panel B). [Mincer-Zarnowitz](#) regressions test $H_0: \beta_1 = 1$, as in Table 2. The average forecast error tests $H_0: \bar{\varepsilon}_t = 0$, as in Table 3. Error-predictability regressions test $H_0: \beta_1 = 0$, as in Table 4. The horizon is the 6-month spot rate, 6 months from now. The units are annualized percentage points. Panel regressions, in the main sample, include exchange fixed effects, compute a within R^2 , and report standard errors clustered by exchange and date. This sample is the longest available for each exchange. Time-series regressions, in the U.S. sample, report [Newey-West](#) standard errors with $L = \lceil 1.3 \times T^{1/2} \rceil$ lags and fixed- b p -values, following [Lazarus et al. \(2018\)](#). This sample is from 01/1990 to 06/2021. See Appendix D.2 for more details.

PANEL A. RISK PREMIA $\tilde{\mu}_{t+6}^{(6)}$												
Mincer-Zarnowitz					Average Error			Error Predictability				N
β_1	$se(\beta_1)$	p -val	R^2	$\tilde{\varepsilon}_t$	$se(\tilde{\varepsilon}_t)$	p -val	β_1	$se(\beta_1)$	p -val	R^2		
Main Sample												
Baseline Filters/Surface	0.56	0.055	***	0.15	0.17	0.11		-0.16	0.047	***	0.03	2227
Outlier Filter	0.60	0.055	***	0.17	0.18	0.098		-0.20	0.050	***	0.05	2241
Open Interest Filter: After 199601	0.53	0.057	***	0.14	0.16	0.11		-0.18	0.045	***	0.04	2033
Outlier/Open Interest Filter: After 199601	0.56	0.051	***	0.15	0.19	0.097	*	-0.21	0.050	***	0.06	2040
Volatility Surface: After 199601	0.57	0.051	***	0.17	0.087	0.096		-0.21	0.053	***	0.06	2160
SVI Surface: U.S. and SX5E	0.59	0.041	*	0.17	0.067	0.12		-0.18	0.030		0.04	609
U.S. Sample												
Bid-Ask Midpoint	0.67	0.096	***	0.22	0.020	0.15		-0.17	0.067	**	0.03	378
Bid Prices	0.64	0.099	***	0.21	0.033	0.14		-0.16	0.078	*	0.03	378
Ask Prices	0.66	0.089	***	0.23	0.0037	0.15		-0.18	0.060	***	0.04	378

PANEL B. EXPECTED RETURNS $\mu_{t+6}^{(6)}$												
Mincer-Zarnowitz					Average Error			Error Predictability				N
β_1	$se(\beta_1)$	p -val	R^2	ε_t	$se(\varepsilon_t)$	p -val	β_1	$se(\beta_1)$	p -val	R^2		
Main Sample												
Baseline Filters/Surface	0.91	0.061		0.59	-0.27	0.10	**	-0.29	0.048	***	0.10	2227
Outlier Filter	0.92	0.062		0.62	-0.27	0.098	**	-0.34	0.050	***	0.14	2241
Open Interest Filter: After 199601	0.91	0.062		0.59	-0.26	0.10	**	-0.31	0.043	***	0.12	2033
Outlier/Open Interest Filter: After 199601	0.92	0.064		0.62	-0.24	0.096	**	-0.35	0.048	***	0.16	2040
Volatility Surface: After 199601	0.93	0.065		0.63	-0.34	0.098	***	-0.36	0.052	***	0.16	2160
SVI Surface: U.S. and SX5E	0.89	0.044		0.65	-0.34	0.12		-0.27	0.052		0.09	609
U.S. Sample												
Bid-Ask Midpoint	0.88	0.073		0.71	-0.40	0.17	**	-0.20	0.092	**	0.04	378
Bid Prices	0.87	0.070	*	0.73	-0.38	0.17	**	-0.19	0.11		0.04	378
Ask Prices	0.87	0.076		0.69	-0.41	0.17	**	-0.21	0.078	**	0.05	378

Table A7
Split-Sample Regressions

This table reports estimates in split samples for option-based risk premia (Panel A) and expected returns (Panel B). [Mincer-Zarnowitz](#) regressions test $H_0: \beta_1 = 1$, as in Table 2. The average forecast error tests $H_0: \bar{\varepsilon}_t = 0$, as in Table 3. Error-predictability regressions test $H_0: \beta_1 = 0$, as in Table 4. The horizon is the 6-month spot rate, 6 months from now. The units are annualized percentage points. Each regression reports [Newey-West](#) standard errors with $L = \lceil 1.3 \times T^{1/2} \rceil$ lags and fixed- b p -values, following [Lazarus et al. \(2018\)](#). See Appendix D.3 for more details.

PANEL A. RISK PREMIA $\tilde{\mu}_{t+6}^{(6)}$												
Mincer-Zarnowitz					Average Error			Error Predictability				N
β_1	$se(\beta_1)$	p -val	R^2	$\tilde{\varepsilon}_t$	$se(\tilde{\varepsilon}_t)$	p -val	β_1	$se(\beta_1)$	p -val	R^2		
U.S. Sample												
Baseline: 199001 to 202106	0.67	0.096	***	0.22	0.020	0.15		-0.17	0.067	**	0.03	378
First Half: 199001 to 200509	0.82	0.12		0.38	0.025	0.15		-0.055	0.074		0.00	189
Second Half: 200510 to 202106	0.56	0.11	***	0.14	0.014	0.24		-0.22	0.075	**	0.05	189
Second Half: 200907 to 201907	0.50	0.10	***	0.20	-0.40	0.12	***	-0.37	0.088	***	0.16	121
PANEL B. EXPECTED RETURNS $\mu_{t+6}^{(6)}$												
Mincer-Zarnowitz					Average Error			Error Predictability				N
β_1	$se(\beta_1)$	p -val	R^2	ε_t	$se(\varepsilon_t)$	p -val	β_1	$se(\beta_1)$	p -val	R^2		
U.S. Sample												
Baseline: 199001 to 202106	0.88	0.073		0.71	-0.40	0.17	**	-0.20	0.092	**	0.04	378
First Half: 199001 to 200509	0.85	0.081		0.72	-0.58	0.25	**	0.022	0.12		-0.00	189
Second Half: 200510 to 202106	0.91	0.14		0.48	-0.22	0.18		-0.32	0.080	***	0.13	189
Second Half: 200907 to 201907	0.58	0.20	*	0.23	-0.50	0.13	***	-0.47	0.074	***	0.23	121

Table A8
Short-Horizon Regressions

This table reports estimates at short horizons for option-based risk premia (Panel A) and expected returns (Panel B). [Mincer-Zarnowitz](#) regressions test $H_0: \beta_1 = 1$, as in Table 2. The average forecast error tests $H_0: \tilde{\varepsilon}_t = 0$, as in Table 3. Error-predictability regressions test $H_0: \beta_1 = 0$, as in Table 4. The horizon is the m -month spot rate, n months from now. The units are annualized percentage points. Panel regressions, in the main sample, include exchange fixed effects, compute a within R^2 , and report standard errors clustered by exchange and date. This sample is the longest available for each exchange. Time-series regressions, in the U.S. sample, report [Newey-West](#) standard errors with $L = \lceil 1.3 \times T^{1/2} \rceil$ lags and fixed- b p -values, following [Lazarus et al. \(2018\)](#). This sample is from 01/1990 to 06/2021. See Appendix D.3 for more details.

PANEL A. RISK PREMIA $\tilde{\mu}_{t+n}^{(m)}$												
Mincer-Zarnowitz				Average Error			Error Predictability					
	β_1	$se(\beta_1)$	p -val	R^2	$\tilde{\varepsilon}_t$	$se(\tilde{\varepsilon}_t)$	p -val	β_1	$se(\beta_1)$	p -val	R^2	N
Main Sample												
$n = 1 \quad m = 1$	0.86	0.063	*	0.48	0.082	0.11		-0.17	0.051	***	0.03	2268
$n = 1 \quad m = 2$	0.86	0.055	**	0.53	0.023	0.088		-0.17	0.048	***	0.04	2269
$n = 2 \quad m = 1$	0.72	0.067	***	0.27	0.032	0.13		-0.28	0.067	***	0.05	2269
U.S. Sample												
$n = 1 \quad m = 1$	0.87	0.063	*	0.52	-0.023	0.064		-0.15	0.071	*	0.02	381
$n = 1 \quad m = 2$	0.89	0.048	**	0.60	-0.073	0.050		-0.13	0.051	**	0.03	382
$n = 2 \quad m = 1$	0.73	0.084	***	0.30	-0.14	0.10		-0.27	0.084	***	0.06	382
PANEL B. EXPECTED RETURNS $\mu_{t+n}^{(m)}$												
Mincer-Zarnowitz				Average Error			Error Predictability					
	β_1	$se(\beta_1)$	p -val	R^2	ε_t	$se(\varepsilon_t)$	p -val	β_1	$se(\beta_1)$	p -val	R^2	N
Main Sample												
$n = 1 \quad m = 1$	0.94	0.059		0.67	-0.056	0.11		-0.23	0.053	***	0.05	2268
$n = 1 \quad m = 2$	0.93	0.049		0.73	-0.13	0.087		-0.23	0.049	***	0.07	2269
$n = 2 \quad m = 1$	0.87	0.065	*	0.54	-0.28	0.13	*	-0.40	0.070	***	0.11	2269
U.S. Sample												
$n = 1 \quad m = 1$	0.97	0.024		0.78	-0.16	0.062	**	-0.20	0.054	***	0.04	381
$n = 1 \quad m = 2$	0.97	0.016	*	0.84	-0.21	0.045	***	-0.15	0.041	***	0.04	382
$n = 2 \quad m = 1$	0.94	0.031	*	0.68	-0.42	0.089	***	-0.32	0.064	***	0.08	382

Table A9
Alternative Horizons

This table reports estimates at alternative horizons for option-based risk premia (Panel A) and expected returns (Panel B). [Mincer-Zarnowitz](#) regressions test $H_0: \beta_1 = 1$, as in Table 2. The average forecast error tests $H_0: \bar{\varepsilon}_t = 0$, as in Table 3. Error-predictability regressions test $H_0: \beta_1 = 0$, as in Table 4. The horizon is the m -month spot rate, n months from now. The units are annualized percentage points. All regressions include exchange fixed effects and compute a within R^2 . Standard errors are clustered by exchange and date. The sample is the longest available for each exchange in the main sample. See Appendix [D.3](#) for more details.

PANEL A. RISK PREMIA $\tilde{\mu}_{t+n}^{(m)}$													
Mincer-Zarnowitz						Average Error			Error Predictability				N
β_1	$se(\beta_1)$	p -val	R^2	$\tilde{\varepsilon}_t$	$se(\tilde{\varepsilon}_t)$	p -val	β_1	$se(\beta_1)$	p -val	R^2			
4-Month Horizon: $n + m = 4$													
$n = 1$	$m = 3$	0.91	0.048	*	0.62	0.048	0.074	-0.11	0.040	**	0.03	2269	
$n = 2$	$m = 2$	0.77	0.058	***	0.35	0.060	0.11	-0.20	0.053	***	0.04	2268	
$n = 3$	$m = 1$	0.69	0.066	***	0.20	0.11	0.14	-0.23	0.056	***	0.03	2235	
5-Month Horizon: $n + m = 5$													
$n = 1$	$m = 4$	0.94	0.043		0.67	0.070	0.065	-0.074	0.033	*	0.01	2269	
$n = 2$	$m = 3$	0.83	0.053	**	0.42	0.12	0.10	-0.13	0.043	**	0.02	2268	
$n = 3$	$m = 2$	0.74	0.063	***	0.25	0.17	0.13	-0.15	0.049	**	0.02	2241	
$n = 4$	$m = 1$	0.62	0.078	***	0.13	0.28	0.16	-0.19	0.053	***	0.02	2236	
6-Month Horizon: $n + m = 6$													
$n = 1$	$m = 5$	0.95	0.037		0.69	0.069	0.059	-0.053	0.029	*	0.01	2269	
$n = 2$	$m = 4$	0.86	0.047	**	0.46	0.14	0.092	-0.093	0.036	**	0.01	2268	
$n = 3$	$m = 3$	0.78	0.058	***	0.30	0.21	0.12	-0.11	0.044	**	0.01	2249	
$n = 4$	$m = 2$	0.67	0.072	***	0.17	0.28	0.14	*	-0.15	0.050	**	0.02	2238
$n = 5$	$m = 1$	0.57	0.077	***	0.09	0.33	0.17	*	-0.20	0.058	***	0.02	2228
9-Month Horizon: $n + m = 9$													
$n = 3$	$m = 6$	0.81	0.055	***	0.38	0.15	0.087		-0.065	0.036		0.01	2257
$n = 4$	$m = 5$	0.73	0.066	***	0.26	0.21	0.10	*	-0.10	0.044	**	0.01	2242
$n = 5$	$m = 4$	0.65	0.071	***	0.17	0.26	0.12	*	-0.14	0.049	**	0.02	2232
$n = 6$	$m = 3$	0.55	0.071	***	0.10	0.28	0.14	*	-0.17	0.051	***	0.02	2224
12-Month Horizon: $n + m = 12$													
$n = 3$	$m = 9$	0.81	0.049	***	0.44	0.095	0.070		-0.059	0.036		0.01	2258
$n = 6$	$m = 6$	0.56	0.055	***	0.15	0.17	0.11		-0.16	0.047	***	0.03	2227
$n = 9$	$m = 3$	0.39	0.097	***	0.06	0.25	0.15		-0.24	0.052	***	0.04	2191

(Continued on the next page)

Table A9
Alternative Horizons (Continued)

PANEL B. EXPECTED RETURNS $\mu_{t+n}^{(m)}$													
		Mincer-Zarnowitz			Average Error			Error Predictability				N	
		β_1	$se(\beta_1)$	p -val	R^2	ε_t	$se(\varepsilon_t)$	p -val	β_1	$se(\beta_1)$	p -val	R^2	N
4-Month Horizon: $n + m = 4$													
$n = 1$	$m = 3$	0.96	0.042		0.80	-0.070	0.073		-0.17	0.040	***	0.05	2269
$n = 2$	$m = 2$	0.90	0.057		0.62	-0.20	0.11		-0.30	0.056	***	0.09	2268
$n = 3$	$m = 1$	0.89	0.068		0.50	-0.25	0.14		-0.36	0.060	***	0.08	2235
5-Month Horizon: $n + m = 5$													
$n = 1$	$m = 4$	0.99	0.038		0.84	-0.024	0.063		-0.12	0.034	***	0.04	2269
$n = 2$	$m = 3$	0.95	0.052		0.69	-0.077	0.098		-0.22	0.046	***	0.06	2268
$n = 3$	$m = 2$	0.94	0.063		0.57	-0.13	0.12		-0.27	0.053	***	0.06	2241
$n = 4$	$m = 1$	0.93	0.080		0.45	-0.10	0.15		-0.33	0.057	***	0.06	2236
6-Month Horizon: $n + m = 6$													
$n = 1$	$m = 5$	0.99	0.035		0.86	-0.025	0.057		-0.099	0.029	***	0.03	2269
$n = 2$	$m = 4$	0.97	0.049		0.73	-0.037	0.086		-0.18	0.040	***	0.05	2268
$n = 3$	$m = 3$	0.97	0.057		0.63	-0.039	0.11		-0.22	0.048	***	0.05	2249
$n = 4$	$m = 2$	0.95	0.071		0.52	-0.077	0.13		-0.29	0.052	***	0.06	2238
$n = 5$	$m = 1$	0.93	0.082		0.42	-0.14	0.15		-0.36	0.060	***	0.07	2228
9-Month Horizon: $n + m = 9$													
$n = 3$	$m = 6$	0.97	0.046		0.73	-0.074	0.082		-0.16	0.036	***	0.05	2257
$n = 4$	$m = 5$	0.97	0.055		0.66	-0.080	0.096		-0.22	0.040	***	0.06	2242
$n = 5$	$m = 4$	0.97	0.063		0.59	-0.10	0.11		-0.27	0.046	***	0.07	2232
$n = 6$	$m = 3$	0.94	0.070		0.51	-0.17	0.13		-0.31	0.053	***	0.08	2224
12-Month Horizon: $n + m = 12$													
$n = 3$	$m = 9$	0.96	0.039		0.78	-0.14	0.067	*	-0.15	0.034	***	0.06	2258
$n = 6$	$m = 6$	0.91	0.061		0.59	-0.27	0.10	**	-0.29	0.048	***	0.10	2227
$n = 9$	$m = 3$	0.81	0.077	**	0.41	-0.45	0.15	**	-0.41	0.062	***	0.12	2191

(Continued from the previous page)

Table A10
Alternative Samples

This table reports estimates in alternative samples for option-based risk premia (Panel A) and expected returns (Panel B). Mincer-Zarnowitz regressions test $H_0: \beta_1 = 1$, as in Table 2. The average forecast error tests $H_0: \bar{\varepsilon}_t = 0$, as in Table 3. Error-predictability regressions test $H_0: \beta_1 = 0$, as in Table 4. The horizon is the 1-month spot rate, 2 months from now. The units are annualized percentage points. All regressions include exchange fixed effects and compute a within R^2 . Standard errors are clustered by exchange and date. The sample is the longest available for each exchange. See Appendix D.3 for more details.

PANEL A. RISK PREMIA $\tilde{\mu}_{t+6}^{(6)}$												
	Mincer-Zarnowitz				Average Error			Error Predictability				
	β_1	$se(\beta_1)$	p -val	R^2	$\tilde{\varepsilon}_t$	$se(\tilde{\varepsilon}_t)$	p -val	β_1	$se(\beta_1)$	p -val	R^2	N
Baseline												
Main: 10 Exchanges	0.76	0.060	***	0.30	0.058	0.13		-0.24	0.060	***	0.04	2198
Full: 20 Exchanges	0.76	0.050	***	0.30	0.14	0.13		-0.24	0.050	***	0.04	4033
Eurozone												
Main: 5 Exchanges	0.74	0.079	**	0.28	0.096	0.14		-0.26	0.079	**	0.04	1040
Full: 8 Exchanges	0.73	0.064	***	0.29	0.13	0.15		-0.27	0.064	***	0.05	1568
Europe												
Main: 7 Exchanges	0.74	0.070	***	0.28	0.074	0.13		-0.26	0.070	***	0.05	1510
Full: 13 Exchanges	0.72	0.055	***	0.29	0.093	0.13		-0.28	0.055	***	0.06	2619
Asia-Pacific												
Main: 2 Exchanges	0.88	0.12		0.36	0.21	0.15		-0.12	0.12		0.01	319
Full: 6 Exchanges	0.86	0.078		0.33	0.35	0.17	*	-0.14	0.078		0.01	1045
Excludes U.S.												
Main: 9 Exchanges	0.77	0.064	***	0.30	0.097	0.13		-0.23	0.064	***	0.04	1829
Full: 19 Exchanges	0.76	0.051	***	0.30	0.17	0.14		-0.24	0.051	***	0.04	3664
PANEL B. EXPECTED RETURNS $\mu_{t+6}^{(6)}$												
	Mincer-Zarnowitz				Average Error			Error Predictability				
	β_1	$se(\beta_1)$	p -val	R^2	ε_t	$se(\varepsilon_t)$	p -val	β_1	$se(\beta_1)$	p -val	R^2	N
Baseline												
Main: 10 Exchanges	0.90	0.063		0.57	-0.25	0.13	*	-0.36	0.065	***	0.09	2198
Full: 20 Exchanges	0.89	0.066		0.55	-0.17	0.13		-0.35	0.057	***	0.09	4033
Eurozone												
Main: 5 Exchanges	0.87	0.082		0.53	-0.21	0.15		-0.39	0.089	**	0.10	1040
Full: 8 Exchanges	0.86	0.079		0.53	-0.19	0.15		-0.40	0.073	***	0.11	1568
Europe												
Main: 7 Exchanges	0.87	0.077		0.54	-0.22	0.13		-0.40	0.080	***	0.11	1510
Full: 13 Exchanges	0.86	0.074	*	0.56	-0.22	0.13		-0.42	0.065	***	0.13	2619
Asia-Pacific												
Main: 2 Exchanges	0.96	0.071		0.59	-0.21	0.16		-0.23	0.10		0.03	319
Full: 6 Exchanges	0.95	0.081		0.50	0.031	0.17		-0.22	0.075	**	0.03	1045
Excludes U.S.												
Main: 9 Exchanges	0.89	0.072		0.55	-0.22	0.13		-0.37	0.071	***	0.09	1829
Full: 19 Exchanges	0.88	0.071		0.54	-0.15	0.14		-0.35	0.059	***	0.09	3664

Table A11
Power Utility Regressions

This table reports estimates from the standpoint of an unconstrained power utility investor fully invested in the market for option-based risk premia (Panel A) and expected returns (Panel B). [Mincer-Zarnowitz](#) regressions test $H_0: \beta_1 = 1$, as in Table 2. The average forecast error tests $H_0: \tilde{\varepsilon}_t = 0$, as in Table 3. Error-predictability regressions test $H_0: \beta_1 = 0$, as in Table 4. The horizon is the 6-month spot rate, 6 months from now. The units are annualized percentage points. Panel regressions, in the main sample, include exchange fixed effects, compute a within R^2 , and report standard errors clustered by exchange and date. This sample is the longest available for each exchange. Time-series regressions, in the U.S. sample, report [Newey-West](#) standard errors with $L = \lceil 1.3 \times T^{1/2} \rceil$ lags and fixed- b p -values, following [Lazarus et al. \(2018\)](#). This sample is from 01/1990 to 06/2021. See Appendix F.1 for more details.

PANEL A. RISK PREMIA $\tilde{\mu}_{t+6}^{(6)}$												
Mincer-Zarnowitz				Average Error			Error Predictability				N	
β_1	$se(\beta_1)$	p -val	R^2	$\tilde{\varepsilon}_t$	$se(\tilde{\varepsilon}_t)$	p -val	β_1	$se(\beta_1)$	p -val	R^2		
Main Sample												
$\gamma = 0.75$	0.50	0.058	***	0.14	0.028	0.058		-0.21	0.045	***	0.05	2218
$\gamma = 1.00$	0.56	0.055	***	0.15	0.17	0.11		-0.16	0.046	***	0.03	2220
$\gamma = 1.25$	0.60	0.057	***	0.15	0.36	0.15	**	-0.13	0.050	**	0.02	2209
$\gamma = 1.50$	0.62	0.062	***	0.16	0.60	0.20	**	-0.10	0.054	*	0.01	2203
$\gamma = 1.75$	0.63	0.068	***	0.16	0.86	0.25	***	-0.071	0.057		0.01	2192
$\gamma = 2.00$	0.64	0.075	***	0.15	1.14	0.29	***	-0.044	0.057		0.00	2179
U.S. Sample												
$\gamma = 0.75$	0.60	0.092	***	0.21	-0.041	0.082		-0.22	0.070	***	0.06	377
$\gamma = 1.00$	0.67	0.096	***	0.22	0.020	0.15		-0.17	0.067	**	0.03	378
$\gamma = 1.25$	0.71	0.098	**	0.23	0.12	0.20		-0.13	0.065	*	0.02	377
$\gamma = 1.50$	0.76	0.10	**	0.24	0.25	0.25		-0.10	0.063		0.01	377
$\gamma = 1.75$	0.80	0.10	*	0.25	0.40	0.30		-0.074	0.061		0.01	377
$\gamma = 2.00$	0.84	0.10		0.26	0.57	0.35		-0.051	0.060		0.00	377
$\gamma = 2.50$	0.91	0.10		0.27	0.94	0.44	**	-0.011	0.058		-0.00	377
$\gamma = 3.00$	0.96	0.10		0.28	1.32	0.51	**	0.019	0.056		-0.00	377
PANEL B. EXPECTED RETURNS $\mu_{t+6}^{(6)}$												
Mincer-Zarnowitz				Average Error			Error Predictability				N	
β_1	$se(\beta_1)$	p -val	R^2	ε_t	$se(\varepsilon_t)$	p -val	β_1	$se(\beta_1)$	p -val	R^2		
Main Sample												
$\gamma = 0.75$	0.92	0.043		0.74	-0.42	0.063	***	-0.44	0.056	***	0.18	2218
$\gamma = 1.00$	0.91	0.061		0.59	-0.28	0.10	**	-0.30	0.046	***	0.11	2220
$\gamma = 1.25$	0.90	0.075		0.49	-0.082	0.15		-0.22	0.048	***	0.07	2209
$\gamma = 1.50$	0.89	0.088		0.43	0.15	0.19		-0.17	0.052	***	0.04	2203
$\gamma = 1.75$	0.89	0.099		0.39	0.42	0.24		-0.13	0.056	**	0.02	2192
$\gamma = 2.00$	0.88	0.11		0.36	0.70	0.28	**	-0.093	0.057		0.01	2179
U.S. Sample												
$\gamma = 0.75$	0.88	0.061	*	0.84	-0.46	0.14	***	-0.26	0.13	*	0.05	377
$\gamma = 1.00$	0.88	0.073		0.71	-0.40	0.17	**	-0.20	0.092	**	0.04	378
$\gamma = 1.25$	0.87	0.083		0.61	-0.30	0.21		-0.16	0.078	*	0.03	377
$\gamma = 1.50$	0.87	0.090		0.53	-0.17	0.26		-0.12	0.071		0.02	377
$\gamma = 1.75$	0.88	0.094		0.48	-0.015	0.30		-0.094	0.067		0.01	377
$\gamma = 2.00$	0.90	0.097		0.45	0.15	0.34		-0.069	0.063		0.00	377
$\gamma = 2.50$	0.93	0.098		0.41	0.52	0.42		-0.027	0.059		-0.00	377
$\gamma = 3.00$	0.96	0.098		0.39	0.91	0.50	*	0.0048	0.056		-0.00	377

Table A12
Long-Horizon Forecast Errors

This table reports estimates of forecast error decay for option-based risk premia. The time- t decay $\phi_t^{(n,m)}$ is the ratio of long to short-horizon forecast errors:

$$\mathbb{E}_t \left[\tilde{\varepsilon}_{t+n}^{(m)} \right] = \phi_t^{(n,m)} \mathbb{E}_t \left[\tilde{\varepsilon}_{t+12}^{(m)} \right].$$

The table below reports the median decay $\phi^{(n,m)}$ by horizon:

$$\phi^{(n,m)} = \text{median} \left\{ \left| \phi_t^{(n,m)} \right| \right\}.$$

The predicted forecast error $\mathbb{E}_t[\cdot]$ is from a time-series regression of future realized forecast errors on current forward rates: the predictor is the 6×6 -month forward rate for $n = 12$ and the $n - 12 \times 12$ -month forward rate for $n \geq 24$. The sample begins in 09/2005 and is the longest available for each horizon in the Eurozone (Euro Stoxx 50). See Appendix F.2 for more details.

m -months	n -months						
	12	24	36	48	60	72	84
12	1.00	1.84	1.51	0.72	1.28	1.66	1.58
24	1.00	1.80	1.54	0.85	1.25	1.97	
36	1.00	1.85	1.63	0.84	1.63		
48	1.00	1.97	1.75	1.34			
60	1.00	1.95	1.94				
72	1.00	2.03					
84	1.00						

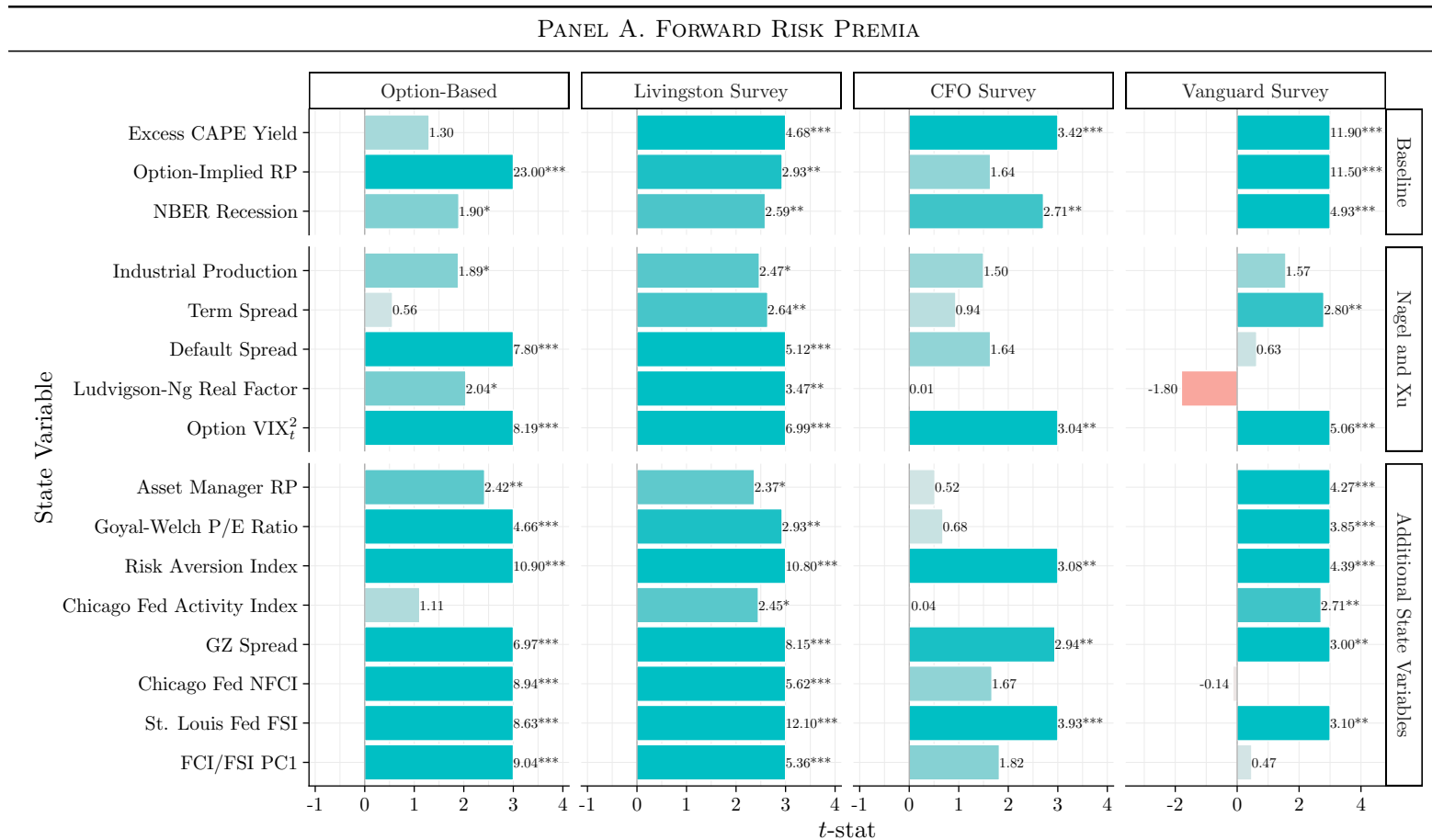
Figure A1
Common Variation in Forward Rates

The left axis plots option-based forward rates. The right axis plots Livingston survey (top panel), CFO survey (middle panel), and Vanguard survey (bottom panel) forward rates. Forward rates are for risk premia. Gray bands are NBER recessions. The Livingston sample is half-yearly from 06/1992 to 06/2021. The CFO sample is quarterly from 12/2001 to 06/2021 and is adjusted for the structural break in 2020. The break adjustment is from a time-series regression of unadjusted forward rates on an indicator variable that equals 1 on and after 09/2020, as in Figure A9. The Vanguard sample is bimonthly from 02/2017 to 06/2021 (including 03/2020). See Section 4.3 for more details.



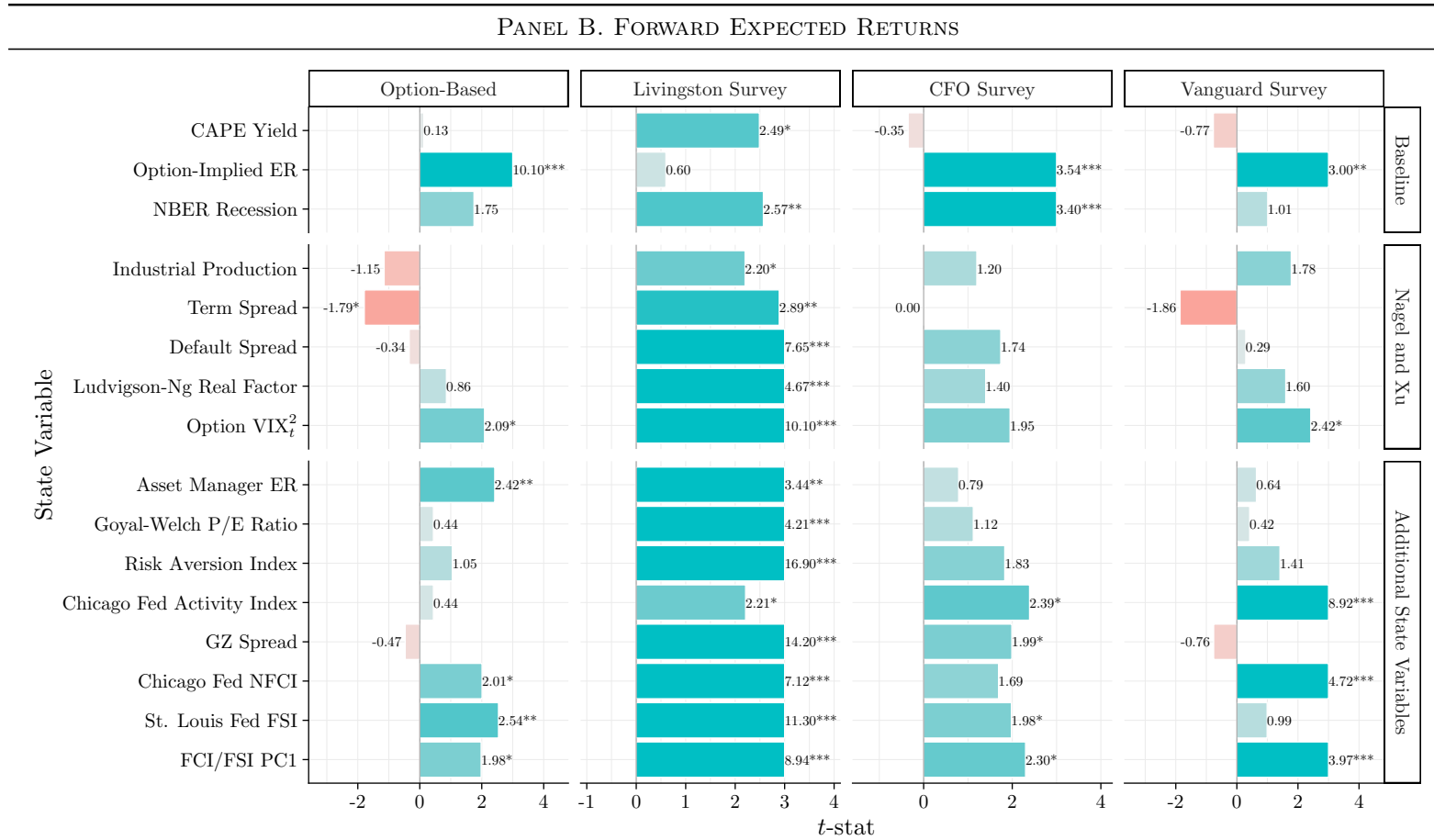
Figure A2
Cyclical Variation in Forward Rates

This figure reports regressions of contemporaneous forward rates on state variables for option-based (first column) and survey-based (second to fourth columns) expectations for risk premia (Panel A) and expected returns (Panel B). The bar length reports the slope t -stat, capped at $|t| = 3.00$. The bar label reports the uncapped t -stat and the statistical significance. On the left-hand side, the option-based horizon is the 6×6 -month forward rate, the Livingston horizon is the 6×6 -month forward rate, the CFO horizon is the 1×9 -year forward rate, and the Vanguard horizon is the 1×9 -year forward rate. On the right-hand side, each state variable is signed such that higher values correspond to bad times (measured as a positive covariance with an NBER recession indicator), so positive signs in the figure indicate countercyclical forward return expectations. Each t -stat is calculated using Newey-West standard errors with $L = \lceil 1.3 \times T^{1/2} \rceil$ lags, following Lazarus et al. (2018), and significance stars are based on two-sided tests with fixed- b critical values (** for 1%, ** for 5%, * for 10%). The option-based sample is monthly from 01/1990 to 12/2021. The Livingston sample is half-yearly from 06/1992 to 12/2021. The CFO sample is quarterly from 12/2001 to 06/2025; each regression includes an indicator variable that equals 1 on and after 09/2020. The Vanguard sample is bimonthly from 02/2017 to 12/2024. See Appendix C.4 for more details.



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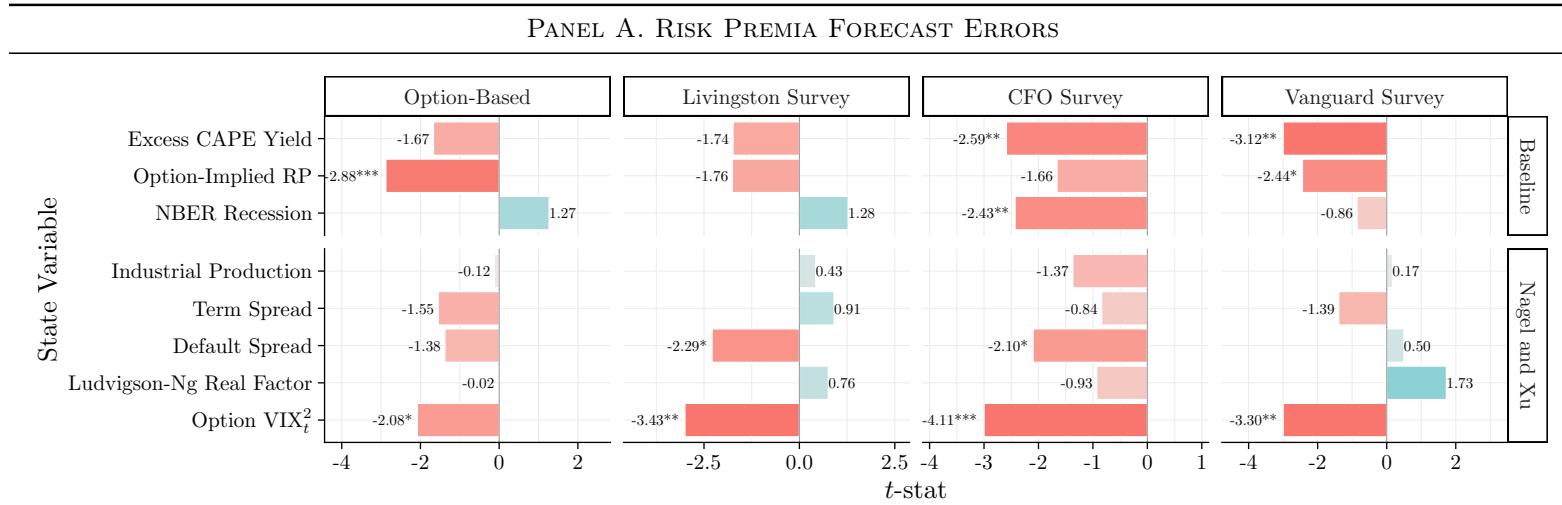
Figure A2
Cyclical Variation in Forward Rates (Continued)



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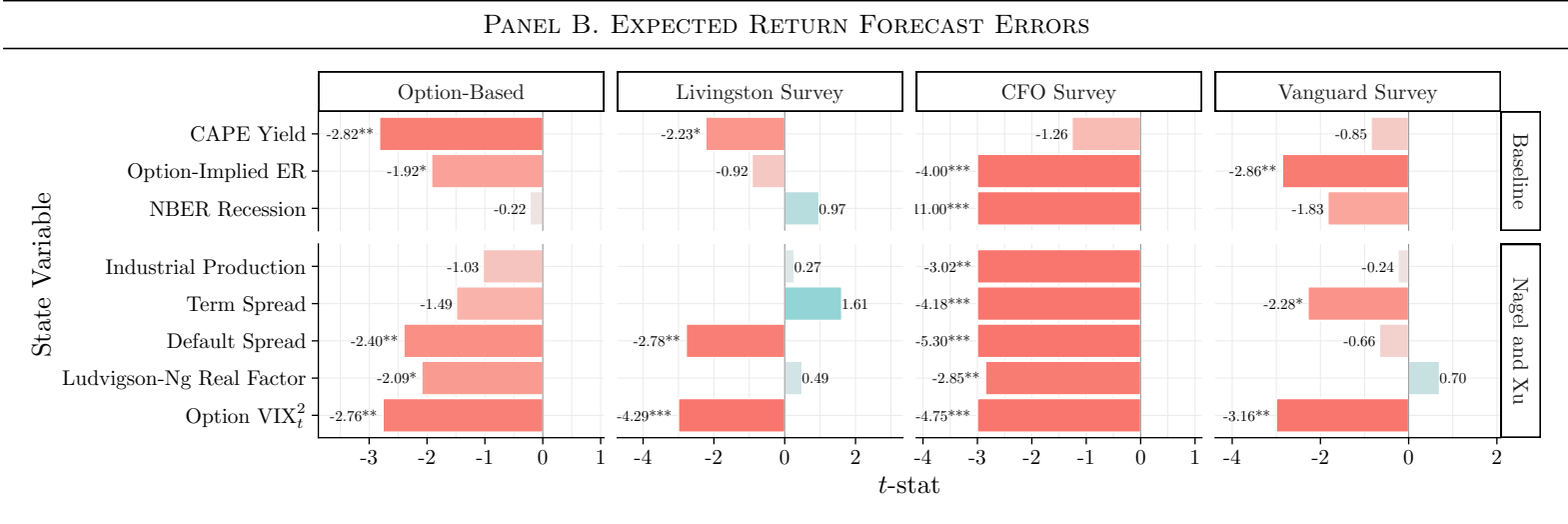
Figure A3
Cyclical Variation in Forecast Errors

This figure reports predictability regressions of future realized forecast errors on current state variables for option-based (first column) and survey-based (second to fourth columns) expectations for risk premia (Panel A) and expected returns (Panel B). The bar length reports the slope t -stat, capped at $|t| = 3.00$. The bar label reports the uncapped t -stat and the statistical significance. On the left-hand side, the option-based horizon is the 6×6 -month forecast error, the Livingston horizon is the 6×6 -month forecast error, the CFO horizon is the 1×9 -year forecast error, and the Vanguard horizon is the 1×9 -year forecast error. On the right-hand side, each state variable is signed such that higher values correspond to bad times (measured as a positive covariance with an NBER recession indicator), so negative signs in the figure indicate procyclical forecast errors. Each t -stat is calculated using Newey-West standard errors with $L = \lceil 1.3 \times T^{1/2} \rceil$ lags, following Lazarus et al. (2018), and significance stars are based on two-sided tests with fixed- b critical values (*** for 1%, ** for 5%, * for 10%). The option-based sample is monthly from 01/1990 to 06/2021 for forward rates and from 07/1990 to 12/2021 for realized spot rates. The Livingston sample is half-yearly from 06/1992 to 06/2021 for forward rates and from 12/1992 to 12/2021 for realized spot rates. The CFO sample is quarterly from 12/2001 to 06/2024 (excluding 09/2019 and 12/2019) for forward rates and from 12/2002 to 06/2025 (excluding 09/2020 and 12/2020) for realized spot rates; each regression includes an indicator variable that equals 1 on and after 09/2020 for forward rates and likewise 09/2021 for realized spot rates. The Vanguard sample is bimonthly from 02/2017 to 12/2023 for forward rates and from 02/2018 to 12/2024 for realized spot rates. See Appendix C.4 for more details.



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Figure A3
Cyclical Variation in Forecast Errors (Continued)



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Figure A4
Price Impact of Option-Based Forecast Errors

This figure plots the option-based estimates for discounted $\mathcal{E}_t$ (red) and the log repurchase-adjusted price-dividend ratio $p_t - d_t$ (blue) in the United States. Discounted forecast errors $\mathcal{E}_t$ are estimated as in (12). The monthly repurchase-adjusted $p_t - d_t$ for the CRSP value-weighted stock index is obtained from [Nagel and Xu \(2022\)](#) via Zhengyang Xu's [website](#). Both series are plotted as log differences from their full-sample means. Gray bands are NBER recessions. The sample is from 01/1990 to 06/2021. See Section 5.1 for more details.

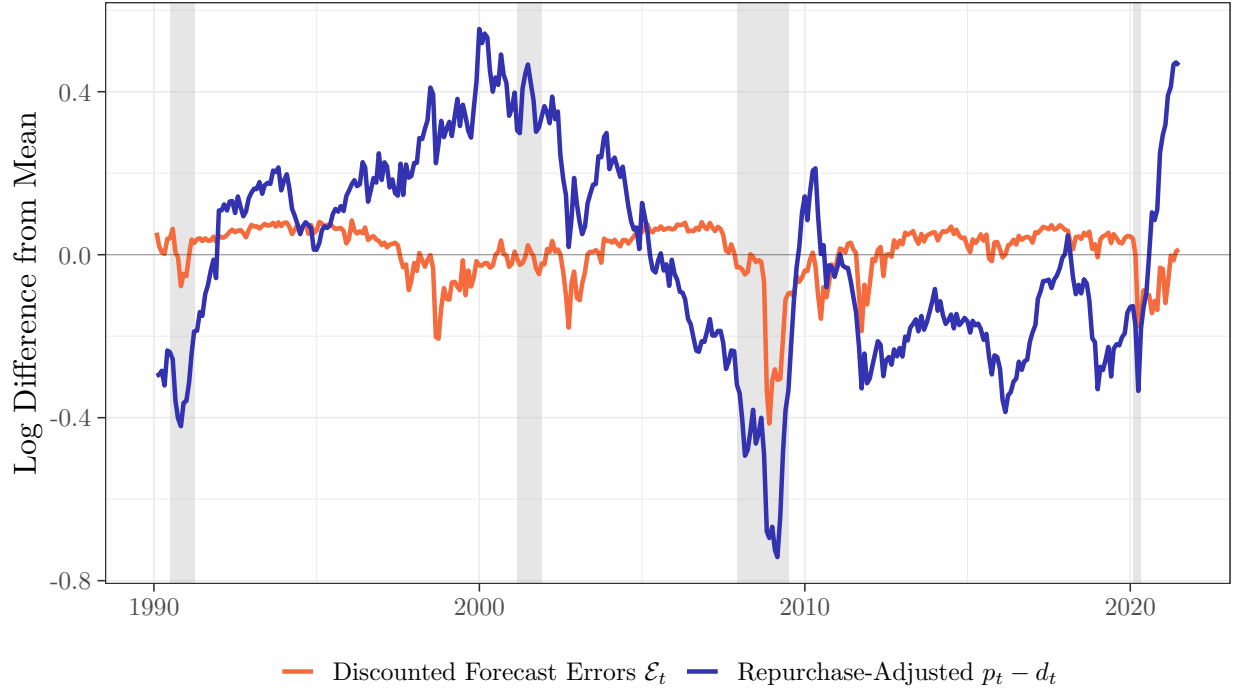


Figure A5
Option-Based Forecast Errors and the Equity Term Structure

This figure plots the effect of predictable risk-premium forecast errors on average realized one-period returns to dividend claims of different maturities. Given a change in expected cash flows $\Delta \mathbb{E}_{t,t+1} d_{t+\nu}$ and a risk-premium forecast error $\tilde{\varepsilon}_{t+1}$ (where the horizon superscript is omitted for simplicity), the log return on a dividend claim with maturity ν is

$$r_{t+1}^{(\nu)} = \mu_t^{(1)} + \Delta \mathbb{E}_{t,t+1} d_{t+\nu} - (\nu - 1) \times \tilde{\varepsilon}_{t+1}.$$

This expression is based on two assumptions: (i) the equity term structure is flat (that is, the expected return on equity claims of different maturities is the same, so the per-period $\mu_t^{(\nu)}$ does not depend on ν); (ii) the term structure of forecast errors is flat (that is, $\tilde{\varepsilon}_{t+1}^{(j)} = \tilde{\varepsilon}_{t+1}^{(j+1)}$ for all j), as in Section 5.1. Fixing cash-flow expectations and holding all else (including risk-free rate expectations) fixed, these assumptions imply that all differences in realized returns arise only from differences in exposure to forecast errors (that is, differences in ν) across claims. We therefore use this equation to calculate the effect of risk-premium forecast errors on realized returns $r_{t+1}^{(\nu)}$ empirically, using realized one-period risk-premium forecast errors from the baseline options sample. Conditional averages are based on the ex ante excess CAPE yield (see Appendix C.4 for details): bad times correspond to the highest 20% of excess CAPE yield values, and good times correspond to the remaining 80% of the sample. See Section 5.2 for more details.

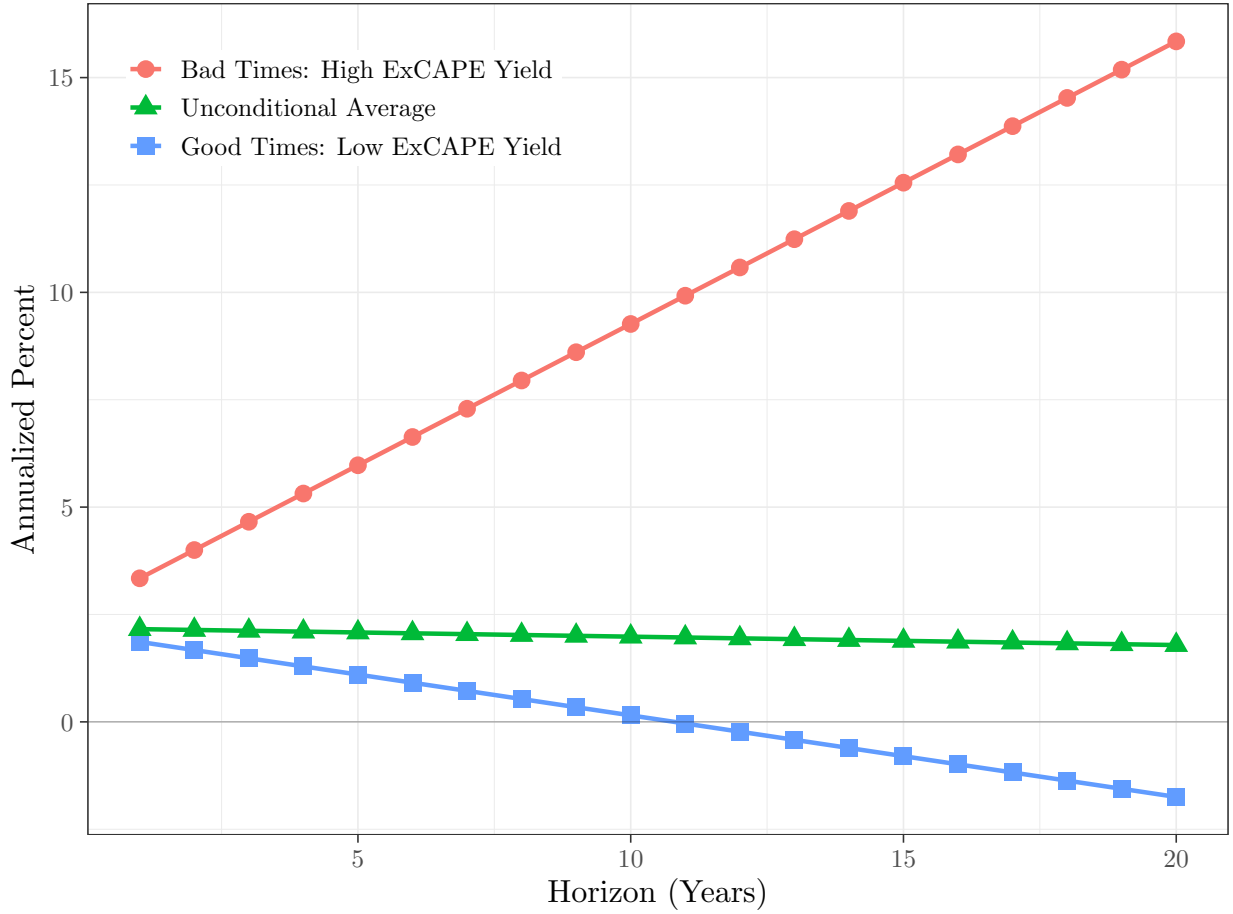


Figure A6
Truncation Bias in Theory

This figure reports truncation bias in the [Black-Scholes](#) (left panel) and [SVJ](#) (right panel) models. Panel A truncates the implied volatility surface at the specified bounds (truncation bounds). Panel B extrapolates the implied volatility surface from the specified bounds (extrapolation bounds). The x-axis units are simple moneyness K/P units from the index price. The y-axis units are non-annualized basis points. The bar units are volatility standard deviations from the forward price: $\frac{K/F-1}{\sigma\sqrt{\tau}}$. Black-Scholes parameters: $P = 100$, $r = 0.05$, $q = 0.00$. SVJ parameters under the risk-neutral measure are from [Bakshi, Cao, and Chen \(1997\)](#):

θ_v	κ_v	σ_v	ρ	μ_J	σ_J	λ
0.040	2.030	0.380	-0.570	-0.050	0.070	0.590

Good times correspond to low volatility ($IV = 0.20$ or $\sqrt{v} = 0.20$), bad times to high volatility ($IV = 0.60$ or $\sqrt{v} = 0.60$). See Appendix [D.1](#) for more details.

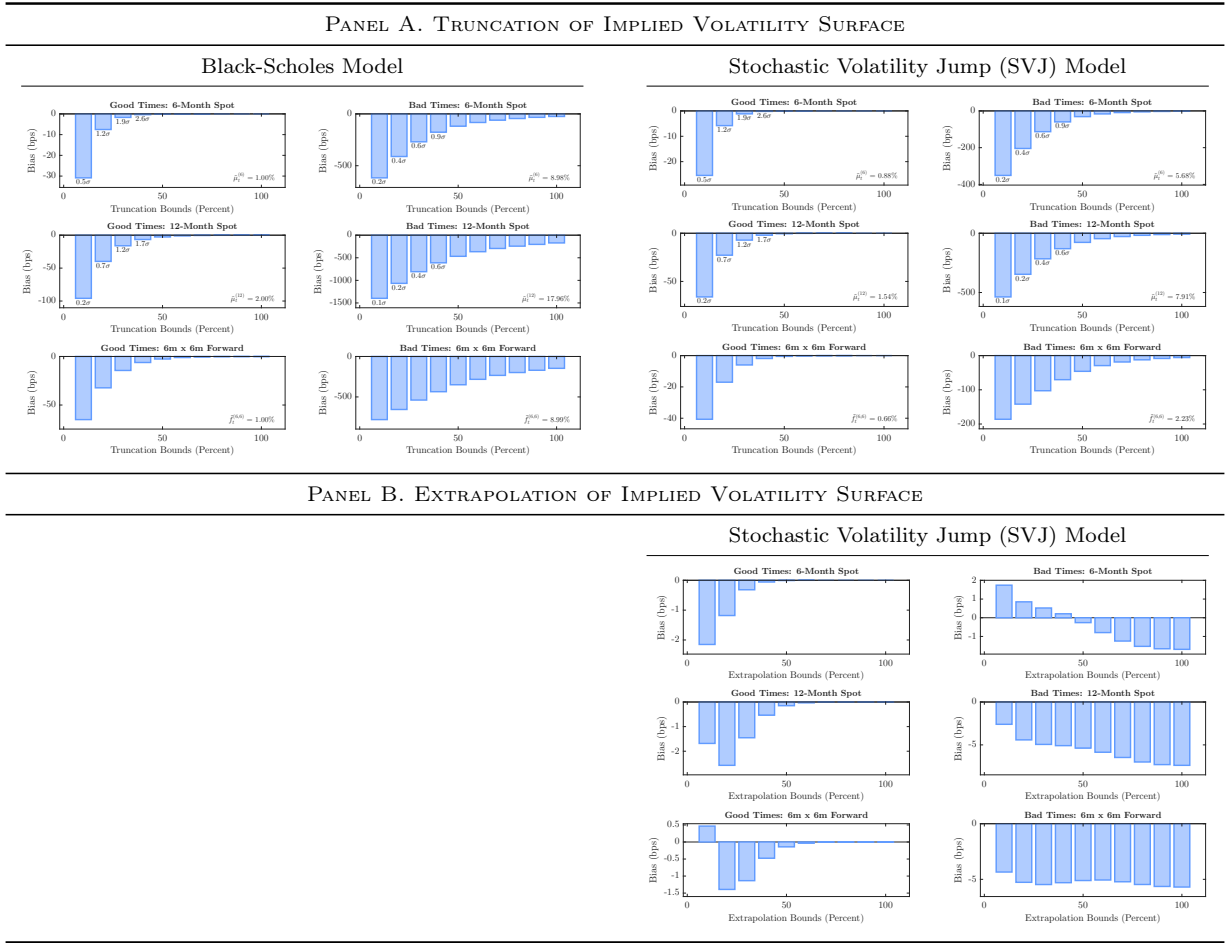


Figure A7
Minimum/Maximum Strike Price: U.S. Sample

This figure plots the minimum/maximum strike price by maturity bin in the U.S. sample. Each bar is the monthly median from daily data. This sample is the longest available for each maturity. The black line is the unconditional median from daily data. This sample is the longest available for all maturities. The minimum is the 1st percentile from out-of-the-money put options. The maximum is the 99th percentile from out-of-the-money call options. The units are risk-neutral standard deviations from the forward price: $\frac{K/F-1}{\sigma\sqrt{\tau}}$. See Appendix D.1 for more details.

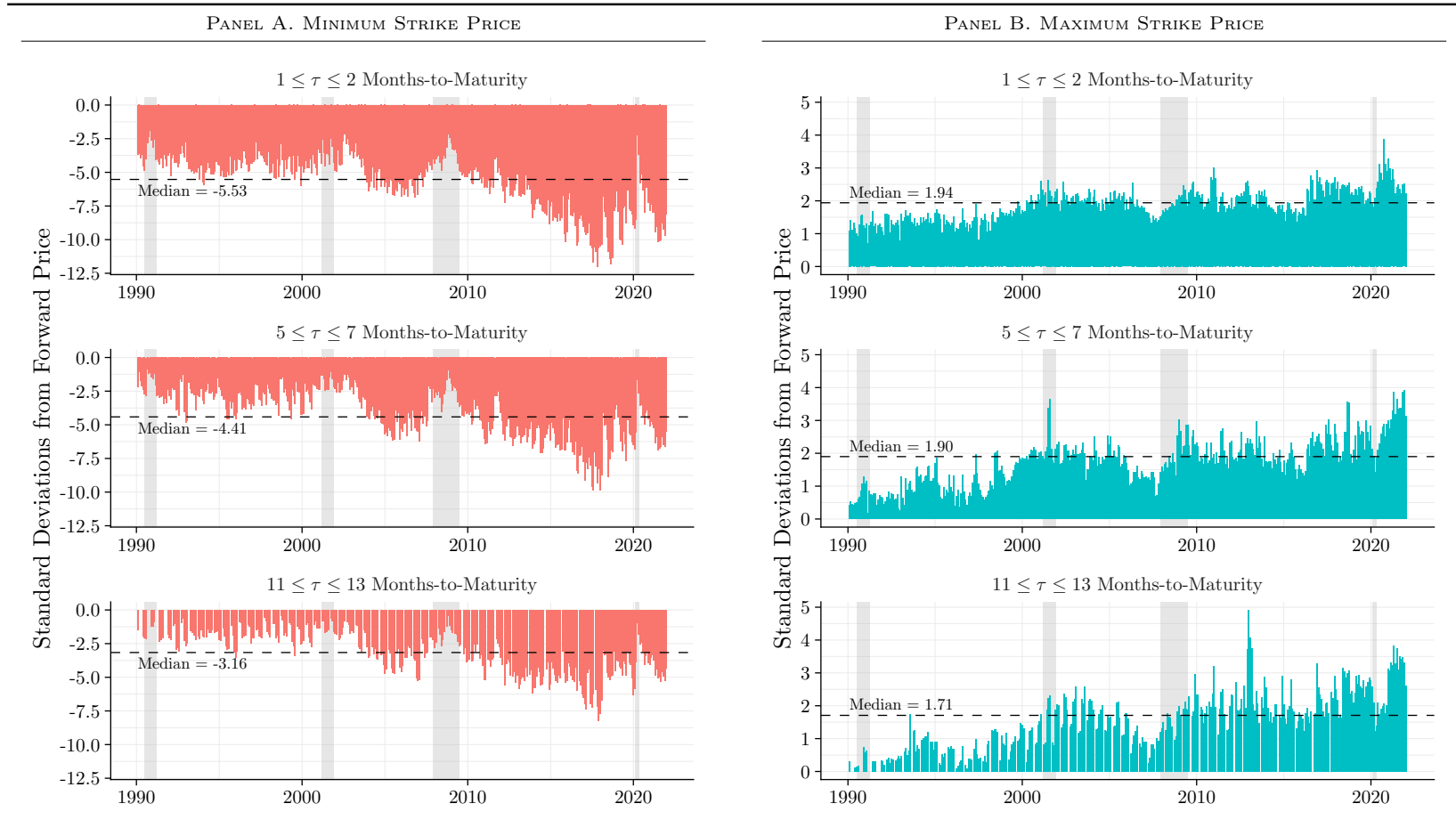


Figure A8
Alternative Integration Bounds

This figure plots 6×6 -month forward rates $\tilde{f}_t^{(6,6)}$ (top panel) and the corresponding realized 6-month spot rates $\tilde{\mu}_{t+6}^{(6)}$ (bottom panel) with alternative integration bounds. Spot and forward rates are for option-based risk premia. Gray bands are NBER recessions. The sample is from 01/1990 to 06/2021 in the U.S. See Appendix D.1 for more details.

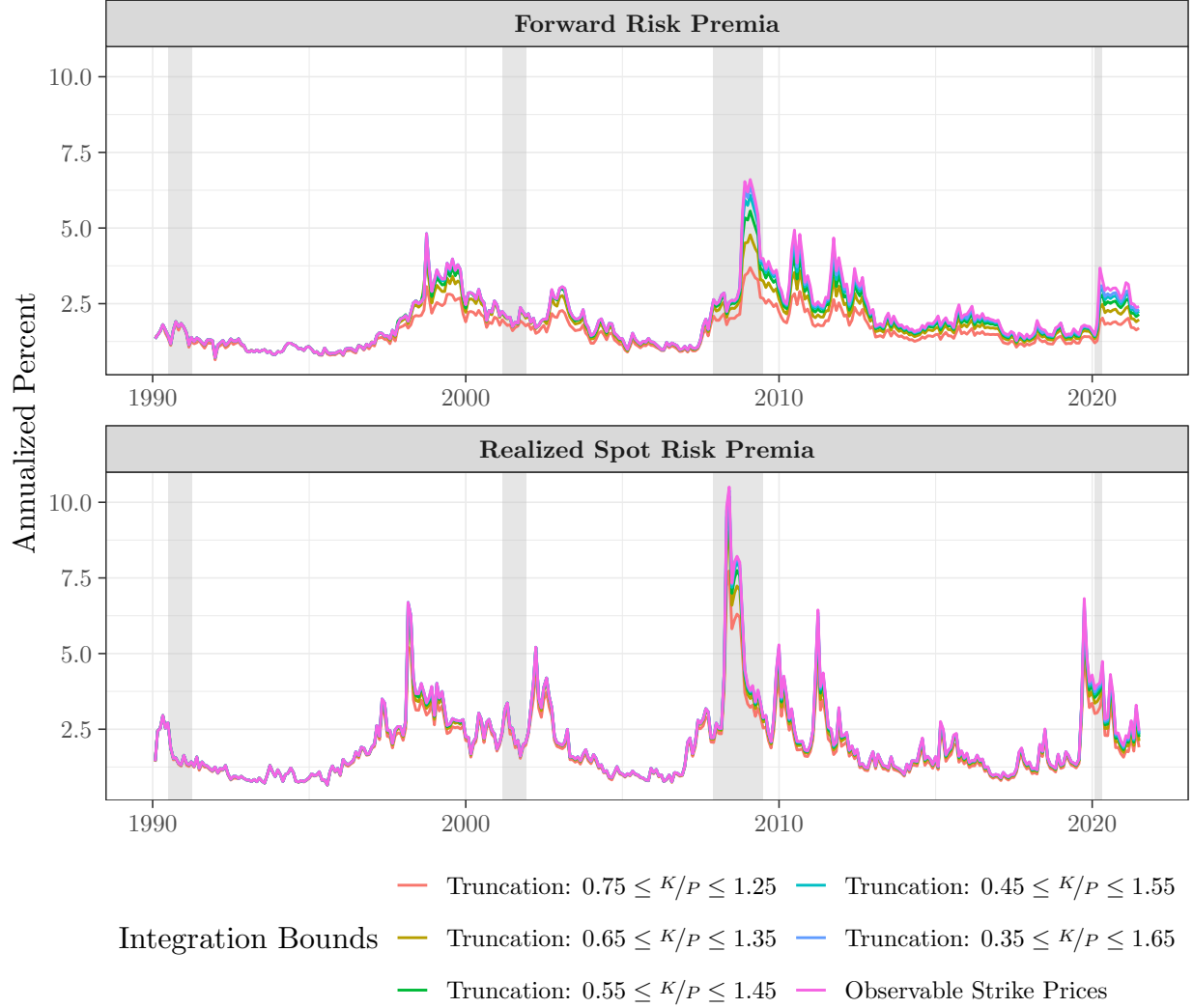


Figure A9
CFO Forward Rates

This figure plots unadjusted 1×9 -year forward rates (top panel) and the corresponding break-adjusted series (bottom panel). The break adjustment is from a time-series regression of unadjusted forward rates on an indicator variable that equals 1 on and after 09/2020:

$$\begin{aligned} \text{unadjusted } \tilde{f}_t^{(1y,9y)} &= \beta_0 + \beta_1 \mathbb{1}_{t \geq 2020} + e_t \\ \text{break-adjusted } \tilde{f}_t^{(1y,9y)} &= \text{unadjusted } \tilde{f}_t^{(1y,9y)} - \beta_1 \mathbb{1}_{t \geq 2020}. \end{aligned}$$

Forward rates are for risk premia. Gray bands are NBER recessions. The sample is quarterly from 12/2001 to 06/2025. See Appendix C.3 for more details.

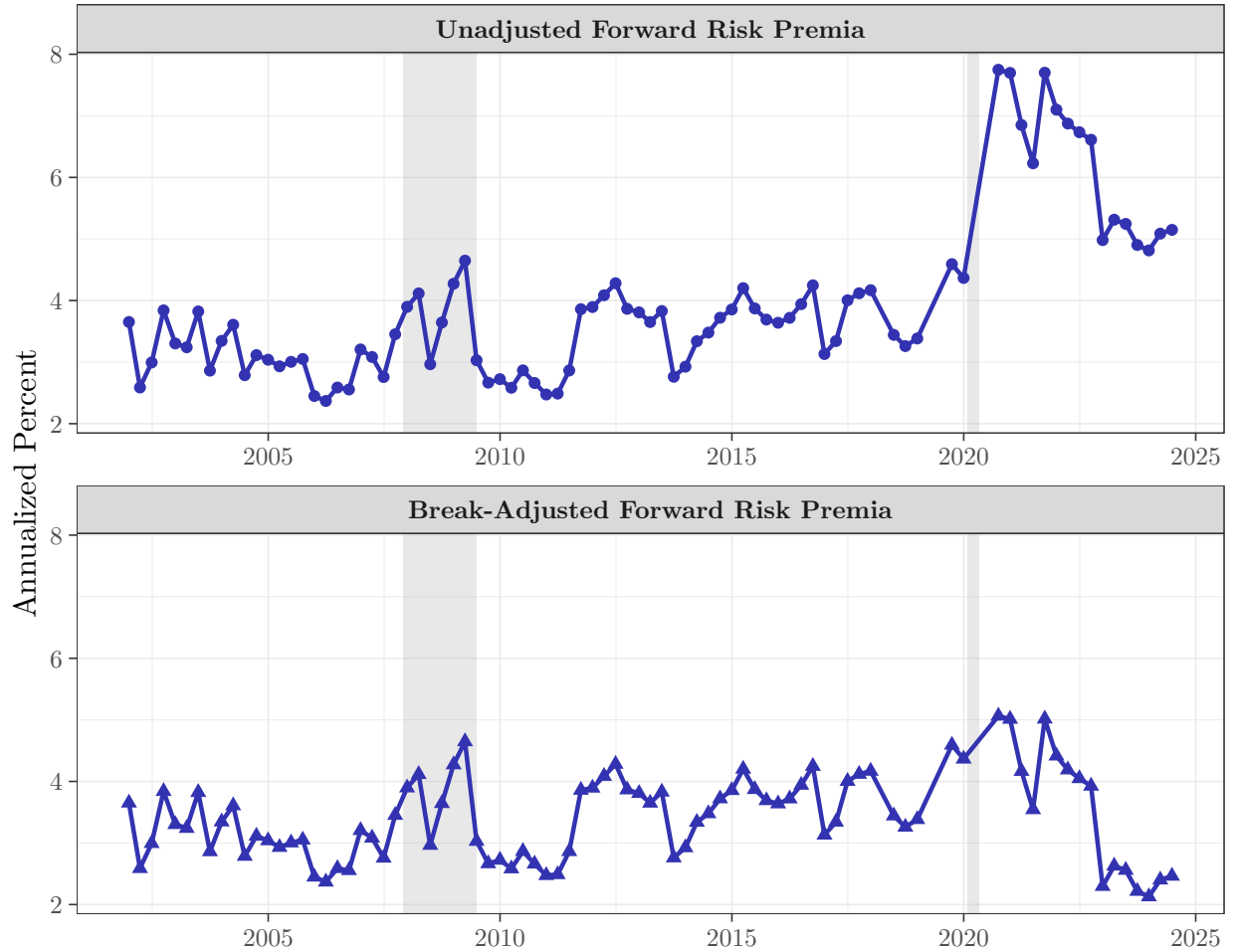


Figure A10
Option-Based Forward Rates and Predicted Forecast Errors

This figure plots instrumented 6×6 -month forward rates $\tilde{f}_t^{(6,6)}$ (top panel) and the corresponding predicted forecast errors $\tilde{\varepsilon}_{t+6}^{(6)}$ (bottom panel). The instrument is the 2×1 -month forward rate, as in Table A3. The predicted forecast error is from a time-series regression of future realized forecast errors on current forward rates, as in Table 4. Forward rates and forecast errors are for risk premia. Gray bands are NBER recessions. The sample is from 01/1990 to 06/2021 in the U.S.

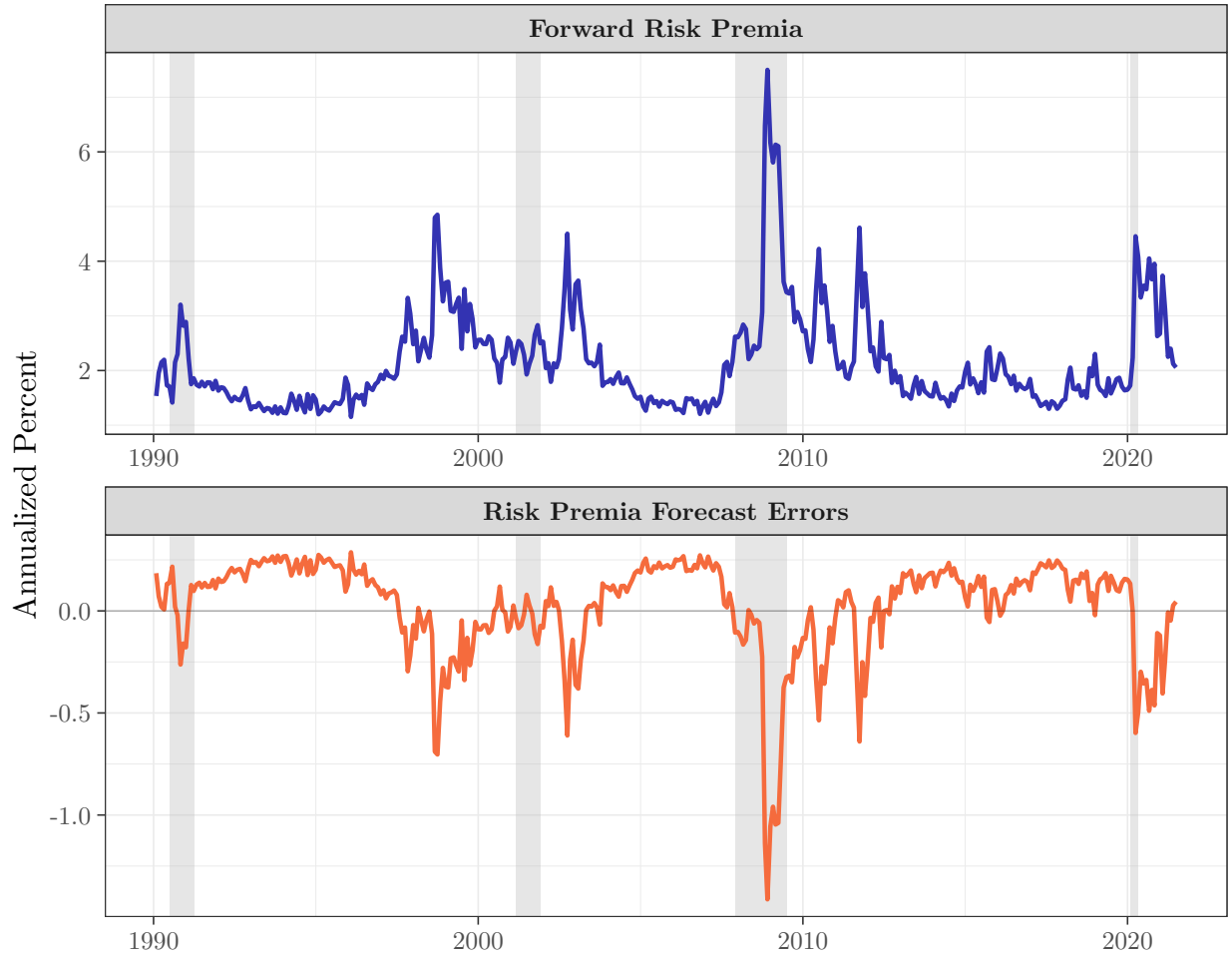


Figure A11
Option-Based Conditional Volatility of the Market Return

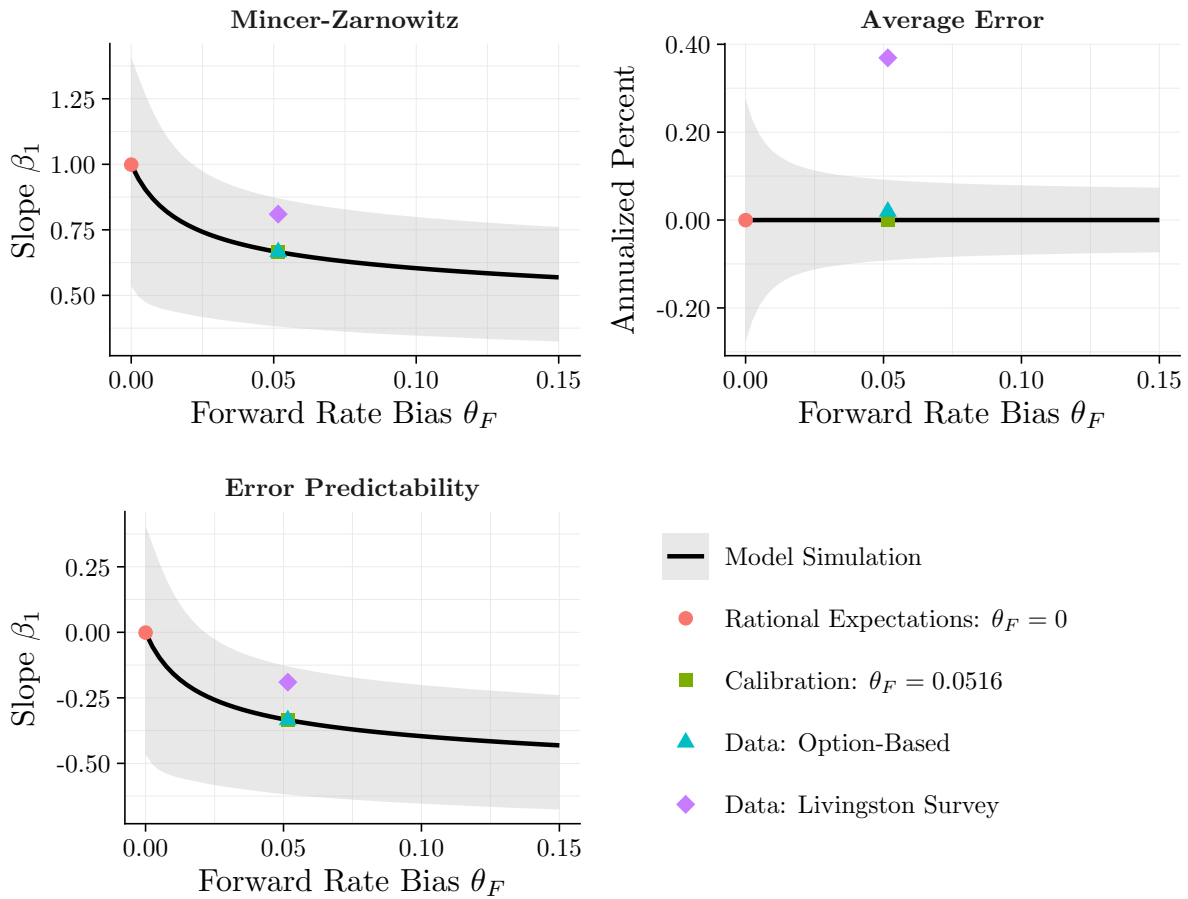
This figure plots the conditional volatility of the 6-month market return $\sigma_t(\ln R_{t,t+6})$ from the standpoint of an unconstrained log utility investor fully invested in the market. Gray bands are NBER recessions. The sample is from 01/1990 to 06/2021 in the U.S. See Appendix G.1 for more details.



Figure A12 Model Calibration

This figure reports estimates in the calibrated model of expectation errors. The model is calibrated with option-based risk premia in the U.S. sample. The black line is the simulated population moment in a single long sample with $T = 37,800,000$ months. The gray ribbon is the simulated 95% confidence interval in 100,000 short samples with $T = 378$ months. The red circle is the simulated moment under rational expectations with $\theta_F = 0$. The green square is the simulated moment under forward rate bias with $\theta_F = 0.0516$. The teal triangle is the observed moment in the data with option-based expectations. The purple diamond is the observed moment in the data with Livingston survey expectations. The horizon is the 6-month spot rate, 6 months from now. The units are annualized percentage points. Short rate bias $\theta_S = 0$ for all simulated moments. See Appendix G.1 for more details.

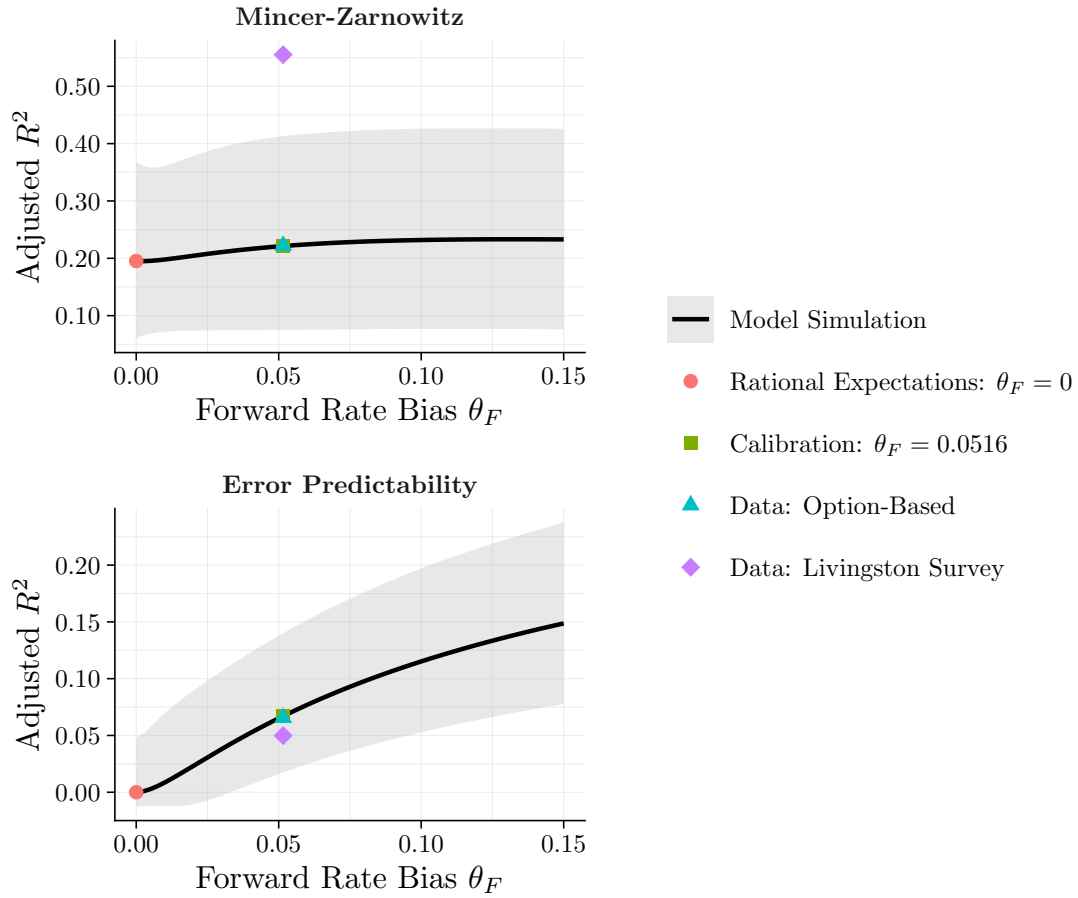
PANEL A. REGRESSION SLOPES AND AVERAGE FORECAST ERRORS



(Continued on the next page)

Figure A12
Model Calibration (Continued)

PANEL B. REGRESSION FIT



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Supplemental Figures

Figure A13
Term Structure of Return Expectations

See Section 2 for more details.

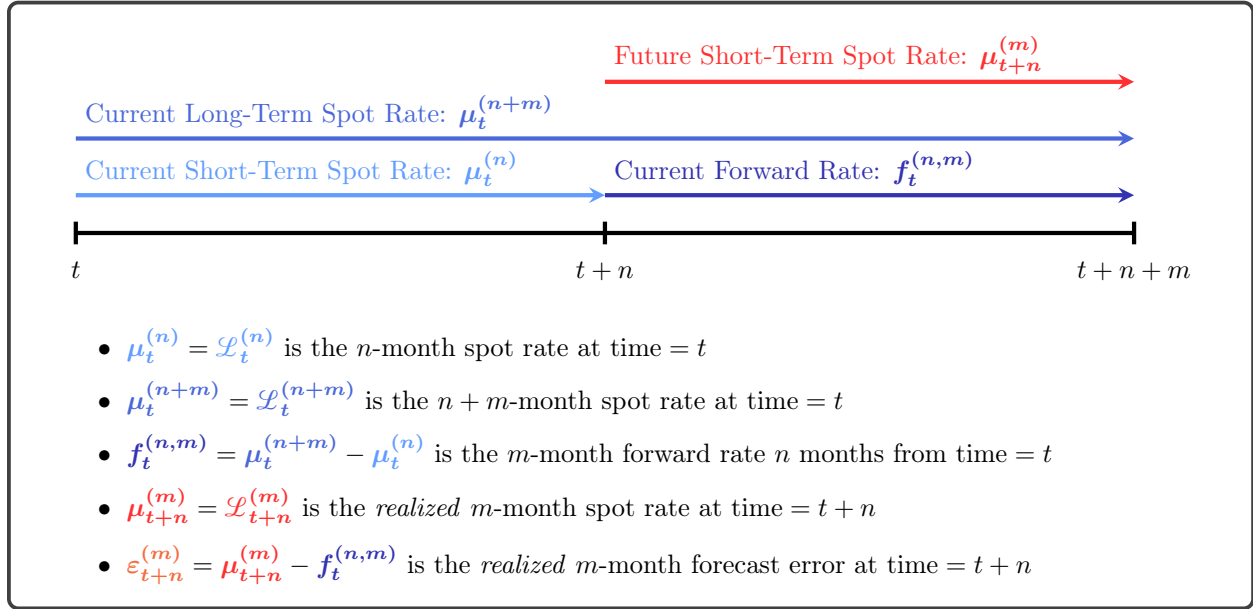


Figure A14 Measuring the LVIX

This figure visualizes the LVIX integral (7) in the [Black-Scholes](#) model:

$$\mathcal{L} - r = \frac{1}{P} \int_0^\infty \omega(K) f(K) dK = \frac{1}{P} \left[\int_0^F \frac{\text{put}(K)}{K} dK + \int_F^\infty \frac{\text{call}(K)}{K} dK \right].$$

Time and time-to-maturity dependence are omitted for simplicity. The x-axis units are volatility standard deviations from the forward price: $\frac{K/F-1}{\sigma\sqrt{\tau}}$. Parameters: $P = 100$, $\Delta K = 0.01$, $r = 0.05$, $\tau = 1$, $IV = 0.20$, $q = 0.00$. See Appendix [D.1](#) for more details.

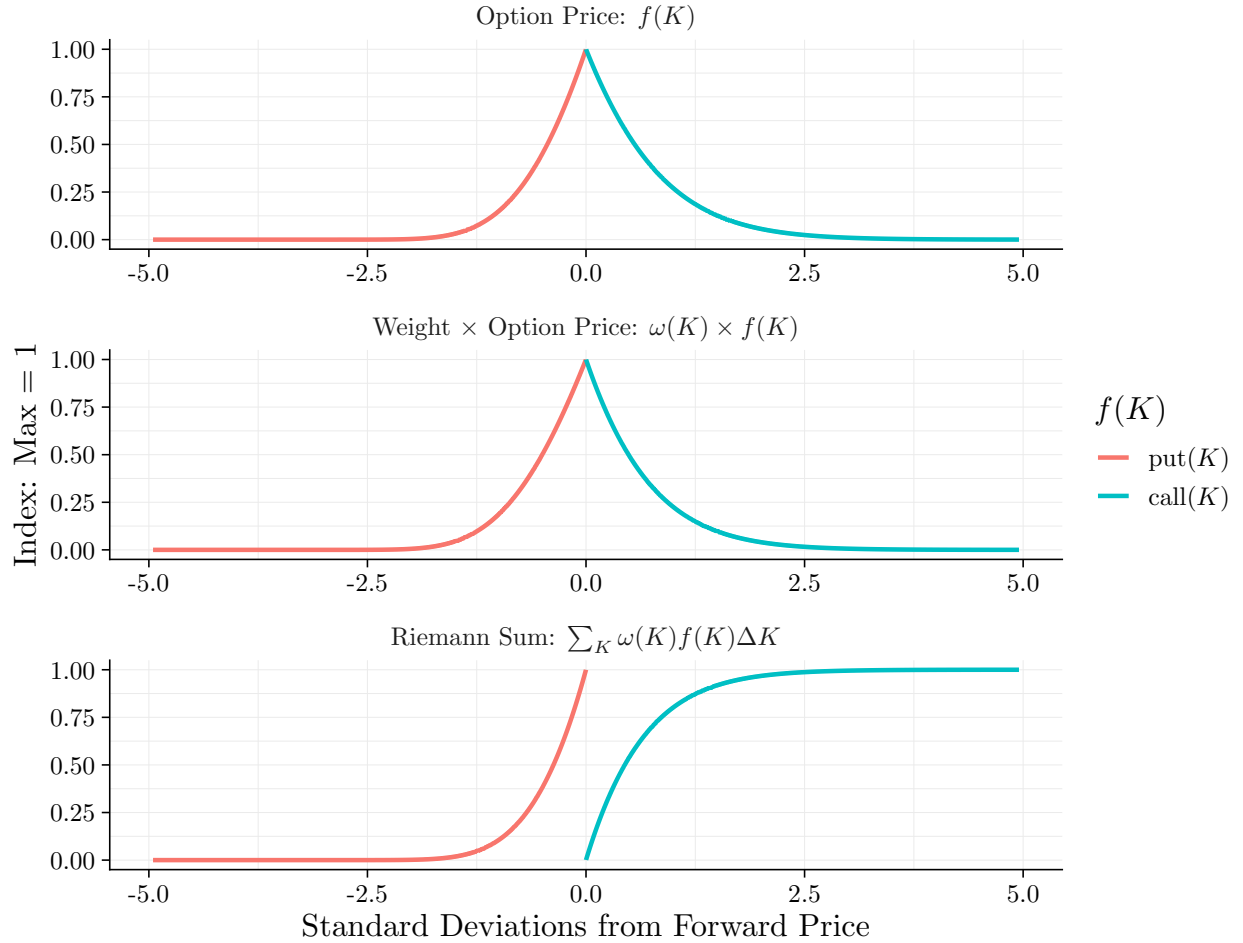


Figure A15
Filtered Option Prices

Panel A plots the number of observations after filters. Panel B plots the share of filtered options with positive open interest. Each bar is the annual median from daily data. This sample is the longest available for each exchange from 1990 to 2020. The black line is the unconditional median from daily data. This sample is from 05/14/2007 to 04/28/2021. Options are out-of-the-money with maturity $7 \leq \tau \leq 365$ days. Belgium, Finland, and the Stoxx Europe 50/600 do not have open interest data. See Appendix C.1 for more details.

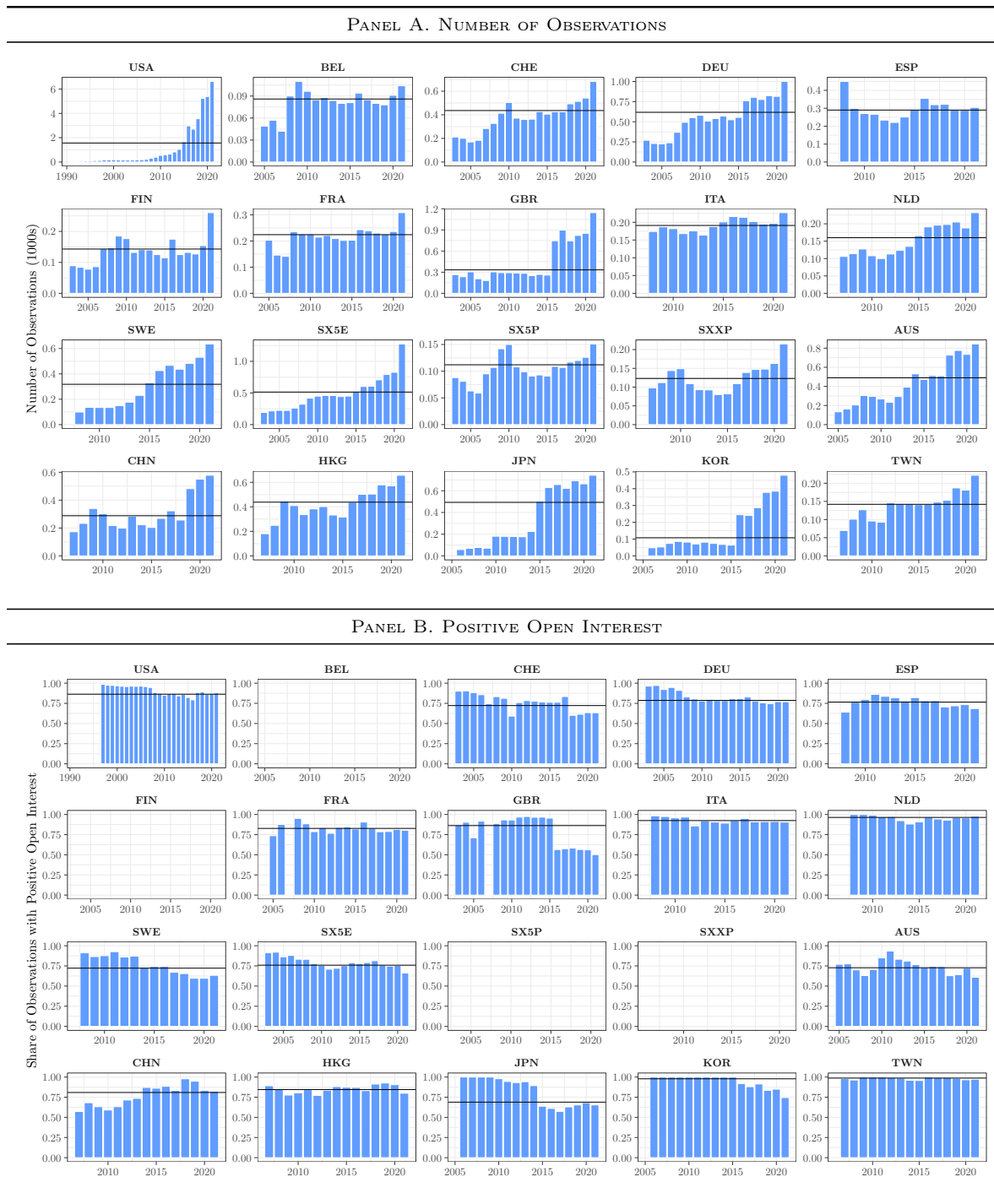


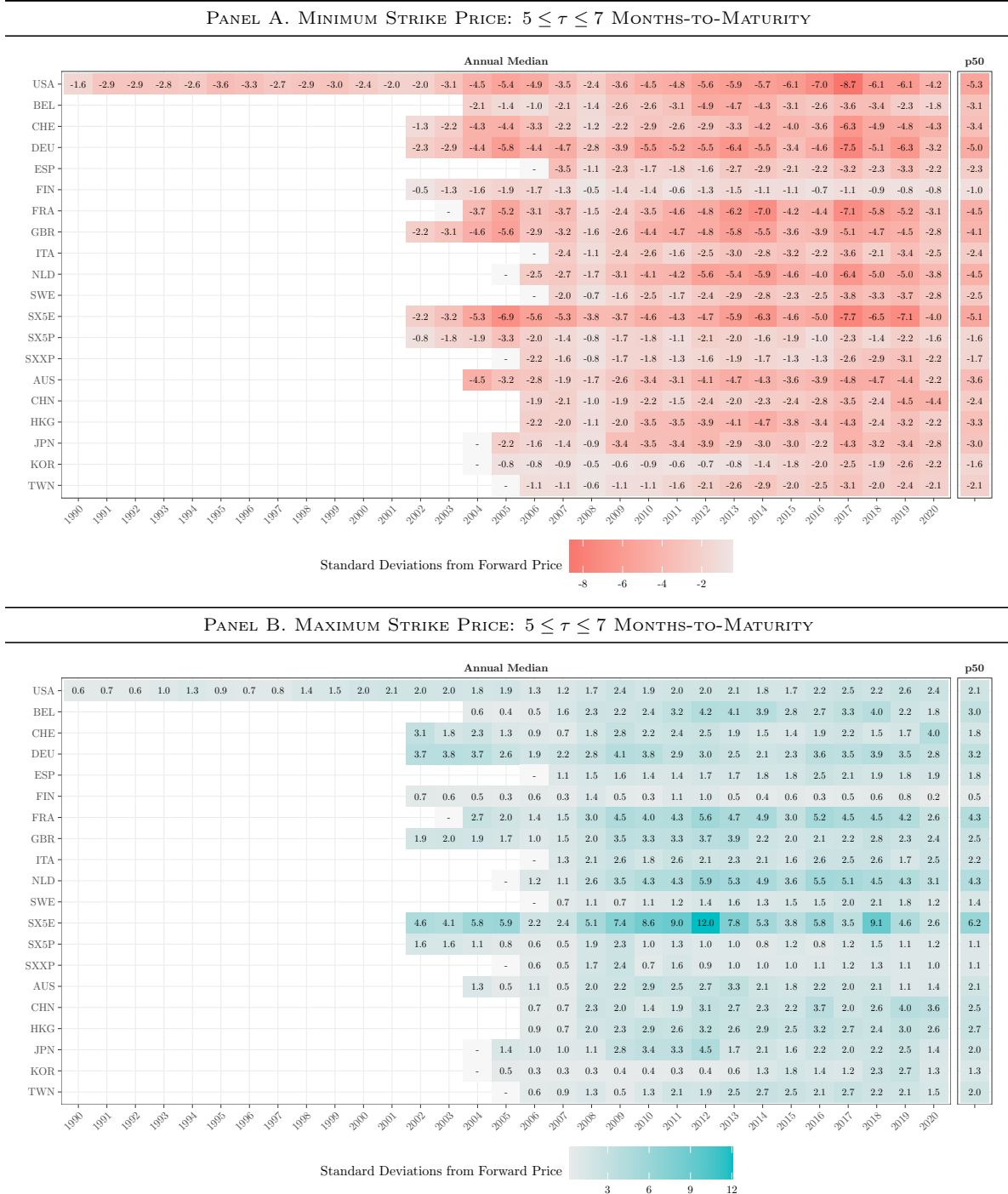
Figure A16
Filtered Option Prices by Maturity

This figure plots the share of filtered options by maturity bin. Each bar is the annual share from daily data. Options are out-of-the-money with maturity $7 \leq \tau \leq 365$ days. The sample is the longest available for each exchange from 1990 to 2020. See Appendix C.1 for more details.



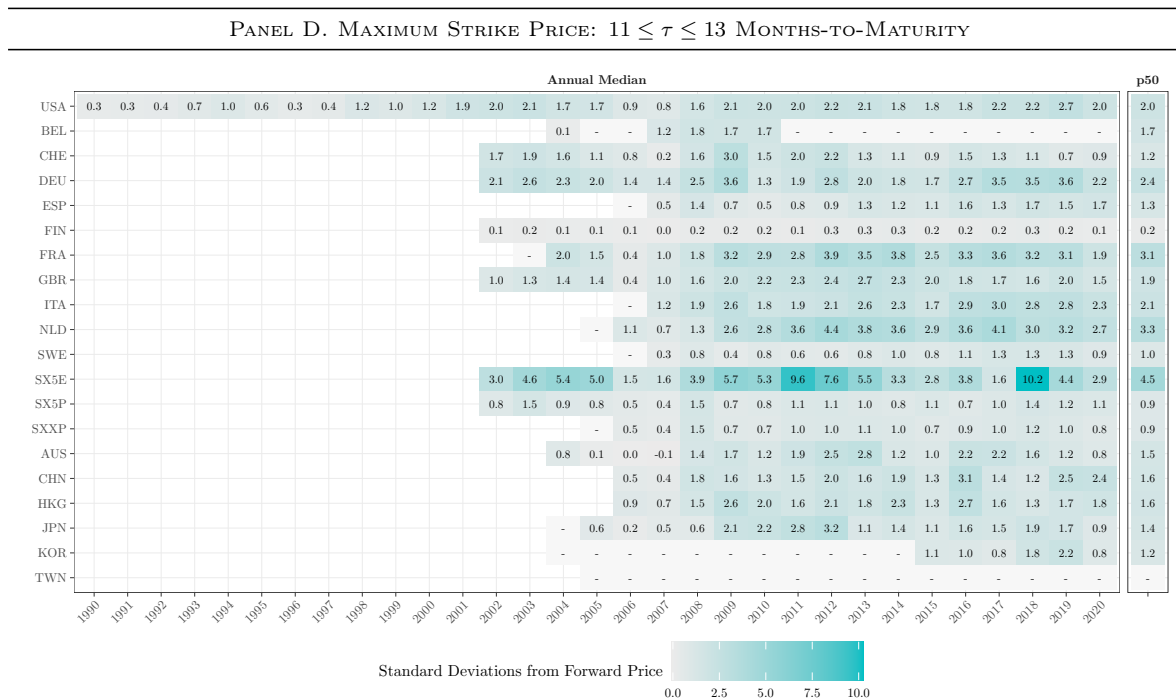
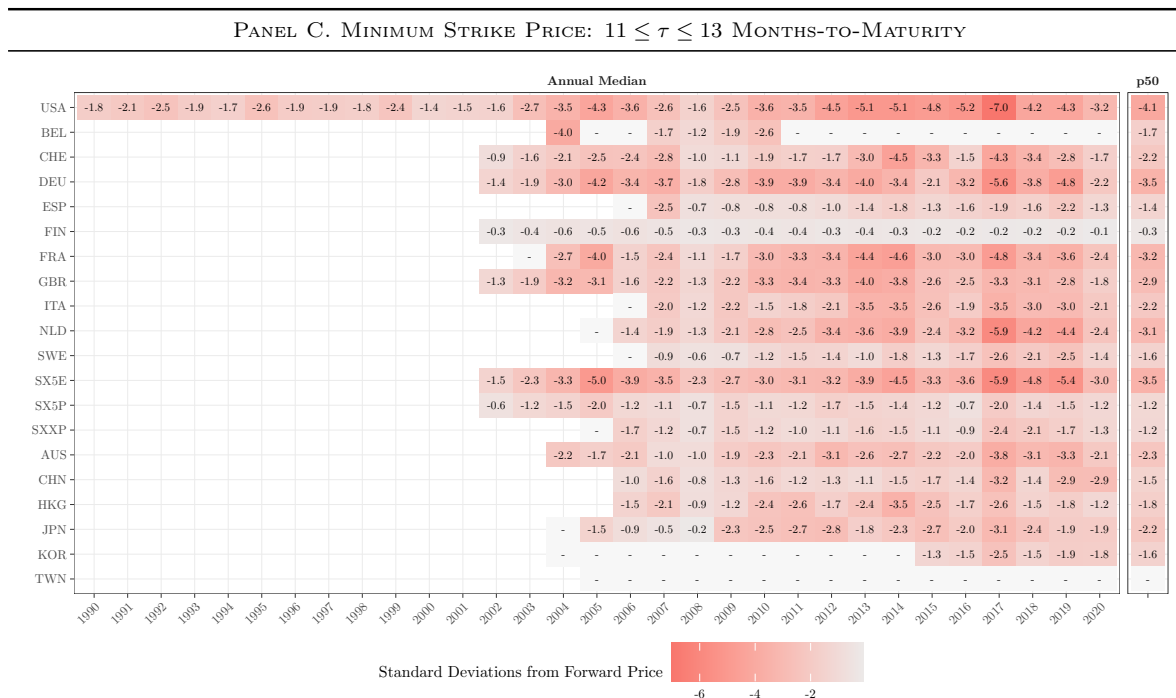
Figure A17
Minimum/Maximum Strike Price: Full Sample

This figure plots the minimum/maximum strike price by maturity bin. The left panel is the annual median from daily data. This sample is the longest available for each exchange from 1990 to 2020. The right panel is the unconditional median from daily data. This sample is from 05/14/2007 to 04/28/2021. The minimum is the 1st percentile from out-of-the-money put options. The maximum is the 99th percentile from out-of-the-money call options. The units are risk-neutral standard deviations from the forward price: $\frac{K/F-1}{\sigma\sqrt{\tau}}$. See Appendix C.1 for more details.



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Figure A17
Minimum/Maximum Strike Price: Full Sample (Continued)



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Figure A18
Surface Triangulation

This figure plots the surface triangulation of option prices on four dates in the U.S. sample. The dots are observed option prices with moneyness $0.80 \leq K/P \leq 1.20$ and maturity $30 \leq \tau \leq 365$ days. See Appendix C.2 for more details.

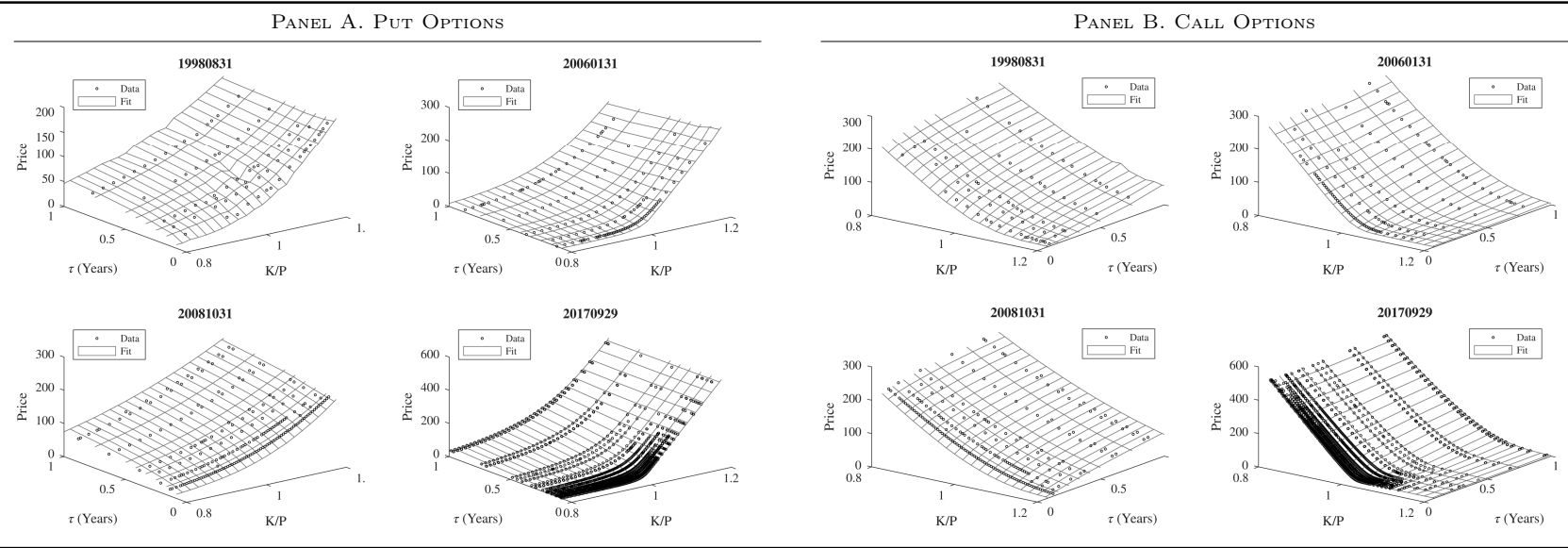


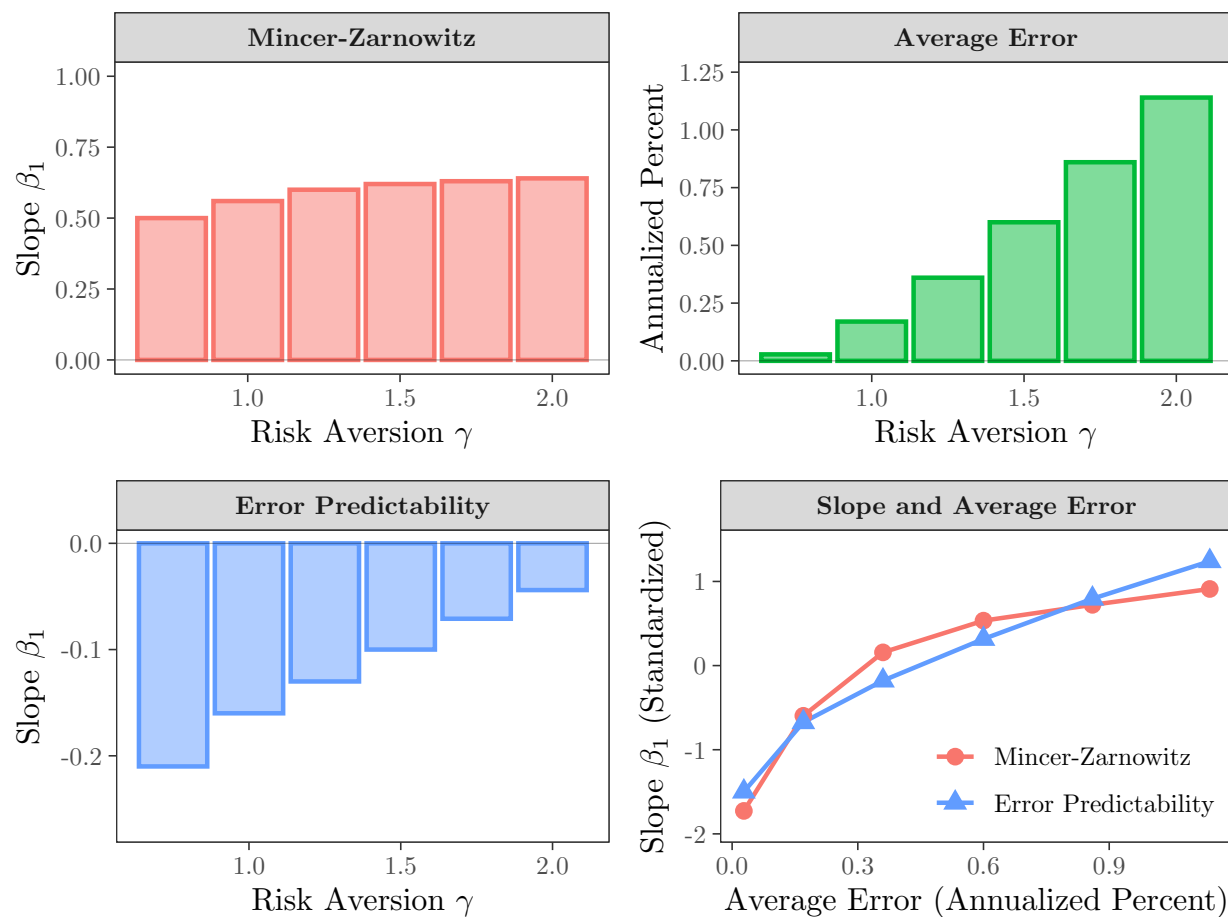
Figure A19
Option-Based Spot and Forward Rates: Full Sample

Panel A plots contemporaneous 6-month spot rates $\tilde{\mu}_t^{(6)}$ (light blue) and 6×6 -month forward rates $\tilde{f}_t^{(6,6)}$ (dark blue). Panel B plots instrumented 6×6 -month forward rates $\tilde{f}_t^{(6,6)}$ (dark blue) and the corresponding realized 6-month spot rates $\tilde{\mu}_{t+6}^{(6)}$ (red). The instrument is the 2×1 -month forward rate: see Table A3 for more details. Spot and forward rates are for option-based risk premia. Gray bands are NBER recessions. The sample is the longest available for each exchange. See Appendix D.3 for more details.



Figure A20
Power Utility: Regression Slopes and Average Forecast Errors

This figure reports estimates from the standpoint of an unconstrained power utility investor fully invested in the market for option-based risk premia. The sample is the longest available for each exchange in the main sample. See Appendix F.1 for more details.



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Alt Text

Figure A1. Three paired time-series charts. Option-based and survey-based forward risk premia co-move, rising together around the 2008 and 2020 recessions.

Figure A2. Horizontal bar charts in four columns report t-statistics from regressions of forward risk premia on sixteen state variables; nearly all estimates are positive, indicating countercyclicality.

Figure A2. A second grid of bar charts for forward expected returns. Most bars remain positive, with fewer statistically significant than for risk premia.

Figure A3. Bar charts of t-statistics from predictive regressions, in the same format. About three quarters of the bars are negative, indicating procyclical forecast errors.

Figure A3. The same format for expected-return forecast errors. Most bars are again negative, most strongly in the CFO column.

Figure A4. A time-series chart. The price-dividend ratio is far more volatile than discounted forecast errors, but both fall sharply in the 2008 recession.

Figure A5. Line charts of calculated returns to dividend claims by maturity: upward-sloping in bad times, downward-sloping in good times, roughly flat on average.

Figure A6. Bar charts of truncation and extrapolation bias. Bias grows sharply negative in bad times as integration bounds narrow, but stays near zero with extrapolation.

Figure A7. Bar charts of monthly median minimum and maximum available strikes. The strike range widens substantially over the sample.

Figure A8. Two stacked time-series charts, six lines each for alternative integration bounds. The lines nearly coincide in calm periods and fan out at crisis peaks.

Figure A9. Two time-series charts of the CFO forward risk premium. After the 2020 break adjustment, the rise in 2020 resembles earlier cyclical peaks rather than a permanent shift.

Figure A10. Two stacked time-series charts. The forward risk premium spikes in crises, and predicted forecast errors turn correspondingly negative.

Figure A11. A time-series chart of option-based conditional volatility, low in calm periods and spiking in stress episodes, most sharply in the 2008 financial crisis.

Figure A12. Three panels of simulated moments against the bias parameter. The calibrated model closely matches the option-based moments; the Livingston moments sit further from the model line.

Figure A12. Two further panels for regression R-squared values. The calibrated model matches the option-based values; one Livingston value lies well above the simulation band.

Figure A13. A schematic timeline with dates t , $t + n$, and $t + n + m$, with arrows marking the spot rates and the forward rate, and definitions beneath.

Figure A14. Three stacked charts of option prices and their weighted counterparts. Nearly all of the integral's mass accrues near the money.

Figure A15. Annual bar charts by exchange of filtered option observations, which grow substantially over time, and of the share with positive open interest.

Figure A16. Stacked annual bar charts by exchange of option observations across four maturity bins. The shortest bin dominates in most markets, with wide variation.

Figure A17. Color-shaded tables of annual median minimum (Panel A) and maximum (Panel B) strike prices, by exchange and year.

Figure A17. The same two tables for options with 11 to 13 months to maturity; the strike range again widens over time.

Figure A18. Three-dimensional wireframe plots for puts and calls on four sample dates. The fitted option-price surfaces track observed prices closely.

Figure A19. Line-chart grids by exchange: spot versus forward risk premia (Panel A), forward versus realized spot premia (Panel B).

Figure A20. Bar charts and a scatter plot of estimates under power utility. As risk aversion increases, regression slopes rise and average forecast errors grow.